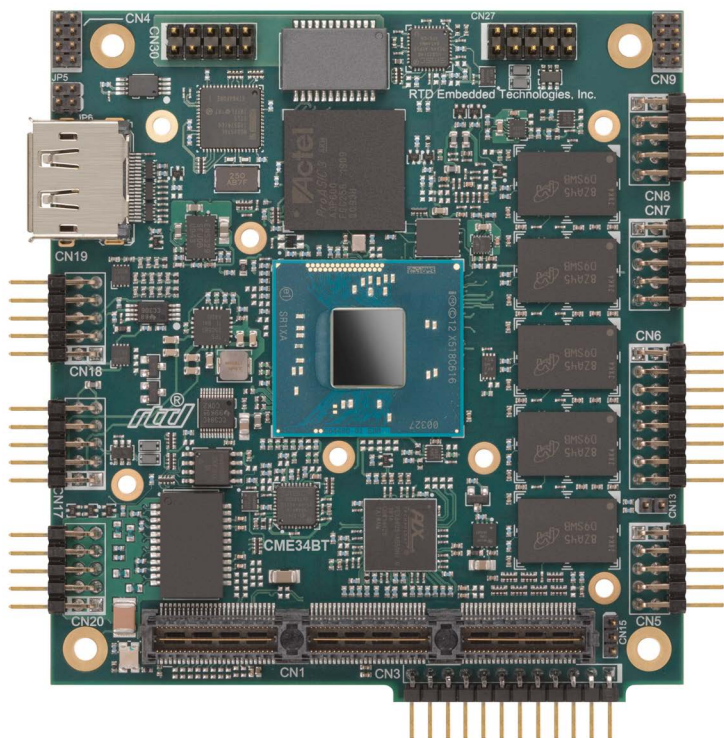


# CME34BT cpuModules™



## User's Manual

BDM-610000088 Revision B



PC/104



# *CME34BT cpuModules™ User's Manual*

RTD Document Number: BDM-610000088 Revision A

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# CME34BT cpuModules™



**PC/104**



## Revision History

| Revision | Date     | Reason for Change              |
|----------|----------|--------------------------------|
| A        | 1/7/2020 | Initial release                |
| B        | 2/2/2023 | Changed fifo size to 512 bytes |

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## **Appendix E Limited Warranty**

# Chapter 1 Introduction

This manual provides comprehensive hardware and software information for users developing with the CME34BT PCIe/104 cpuModule.



**Note** Read the specifications beginning on page 11 prior to designing with the cpuModule.

---

This manual is organized as follows:

|                   |                                                                                                                                                                                                                                                  |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Chapter 1</b>  | <b><i>Introduction</i></b><br>introduces main features and specifications                                                                                                                                                                        |
| <b>Chapter 2</b>  | <b><i>Getting Started</i></b><br>provides abbreviated instructions to get started quickly                                                                                                                                                        |
| <b>Chapter 3</b>  | <b><i>Connecting the cpuModule</i></b><br>provides information on connecting the cpuModule to peripherals                                                                                                                                        |
| <b>Chapter 4</b>  | <b><i>Using the cpuModule</i></b><br>provides information to develop applications for the cpuModule, including general cpuModule information, detailed information on storing both applications and system functions, and using utility programs |
| <b>Appendix A</b> | <b><i>Hardware Reference</i></b><br>lists jumper locations and settings, physical dimensions, and processor thermal management                                                                                                                   |
| <b>Appendix B</b> | <b><i>Troubleshooting</i></b><br>offers advice on debugging problems with your system                                                                                                                                                            |
| <b>Appendix C</b> | <b><i>IDAN™ Dimensions and Pinout</i></b><br>provides connector pinouts for the cpuModule installed in an RTD Intelligent Data Acquisition Node (IDAN) frame                                                                                     |
| <b>Appendix D</b> | <b><i>Additional Information</i></b><br>lists sources and websites to support the cpuModule installation and configuration                                                                                                                       |
| <b>Appendix E</b> | <b><i>Limited Warranty</i></b>                                                                                                                                                                                                                   |

# CME34BT cpuModules

RTD's CME34BT cpuModule represents the latest in low power PCI Express embedded controllers. Based on an Intel Atom® E3800 Series processor and chipset, the CME34BT is offered with either a low power 1.46 Ghz Single-Core processor, a 1.33GHz Dual-Core processor, or a 1.91 GHz Quad-Core processor. The DDR3 memory interface on the single- and dual-core models operate up to 1066 MT/s (533 MHz) while the quad-core is 1333- MT/s (666 MHz). All memory chips are soldered directly onto the board.

The video interface of the cpuModule's Intel Atom's processor supports dual analog SVGA and DisplayPort outputs. The two video outputs are independent, and can display separate images and display timings, with audio support on the DisplayPort. Maximum resolution of both video outputs is 1920 x 1200.

A Serial-ATA (SATA) controller provides links to one disk drive on the PCIe bus connector as well as an onboard solid-state disk drive which is soldered to the board for maximum reliability. Network connectivity is provided by two integrated 10/100/1000 Mbps Ethernet controllers. High-speed peripheral connections include USB 2.0, with up to 480 Mb/sec data throughput. One additional SATA link and two USB 2.0 connections permit further expandability on the top-side PCIe/104 Type 2 connector. The bottom PCIe/104 connector also has two USB 2.0 connections and four x1 PCI Express Gen 2.0 links. Other features include RS-232/422/485 COM ports and Advanced Digital I/O (aDIO).

RTD has gone the extra mile to include additional advanced features for maximum flexibility. These include an onboard surface-mount flash drive with a standard SATA interface. An Advanced Watchdog Timer is provided that can generate an interrupt or reset when the timer expires. The CME34BT is also available in a rugged and fanless IDAN enclosure.

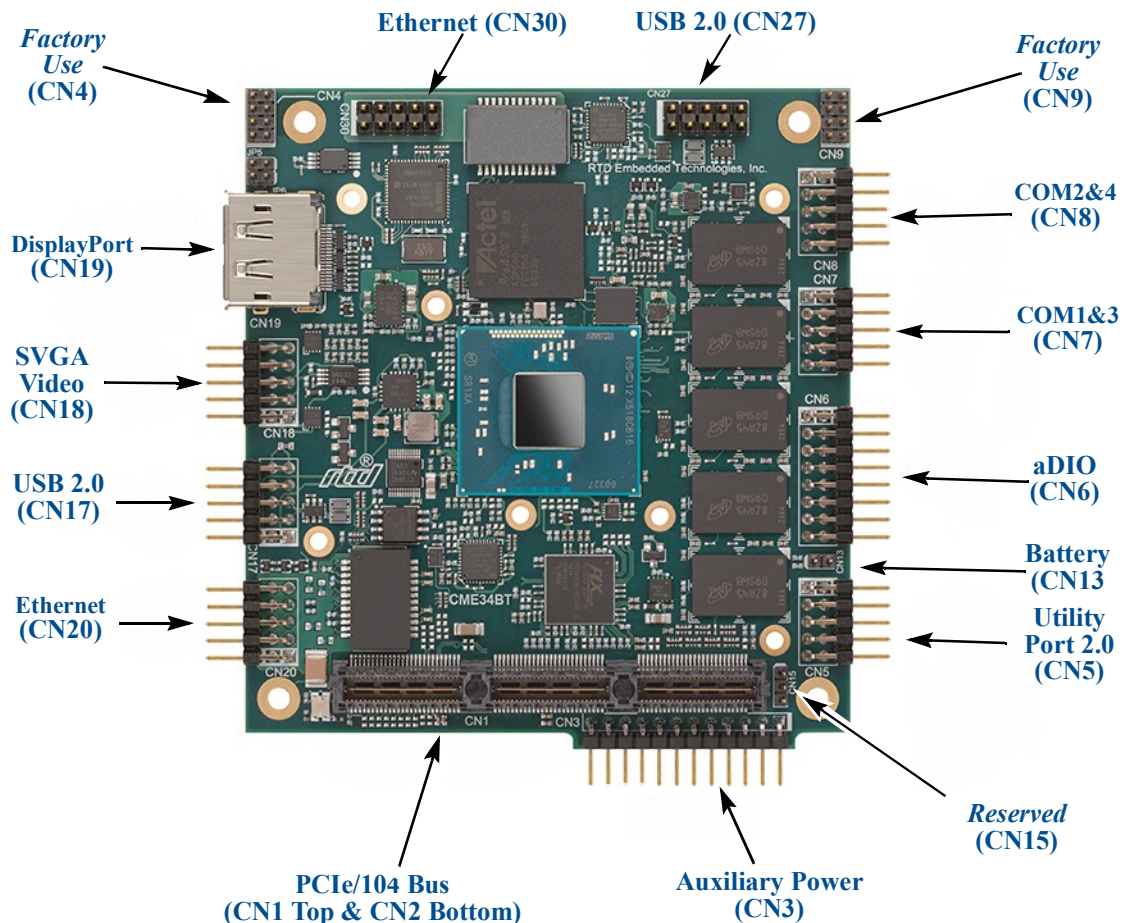


Figure 1 CME34BT cpuModule (top view)

You can easily customize the cpuModule by stacking PCI/104-Express or PCIe/104 modules such as video controllers, Digital Signal Processors, drive carriers, LAN controllers, or analog and digital data acquisition modules. Stacking modules onto the cpuModule avoids expensive installations of backplanes and card cages, and preserves the module's compactness.

The cpuModule uses the RTD Enhanced BIOS. Drivers in the BIOS allow booting from hard disk, or UEFI shell, thus enabling the system to be used with traditional disk drives or nonmechanical drives. Booting to USB devices is also supported.

The cpuModule and BIOS are compatible with any real-time operating systems for PC compatible computers, although these may require creation of custom drivers to use the aDIO and watchdog timer.

## **aDIO**

RTD's exclusive aDIO™ is 12 digital bits configured as 8 bit-direction programmable and 4-bit port-direction programmable I/O, plus 2 strobe inputs giving you any combination of inputs and outputs. Match, event, and strobe interrupt modes mean no more wasting valuable processor time polling digital inputs. Interrupts are generated when the 8 bit-direction programmable digital inputs match a pattern or on any value change event. Bit masking allows selecting any subgroup of eight bits. The strobe input latches data into the bit-programmable port and generates an interrupt.

## Ordering Information

The CME34BT cpuModule is available with a selection of processors and heatsinks. The cpuModule can also be purchased as part of an Intelligent Data Acquisition Node (IDAN™) building block, which consists of the cpuModule and a milled aluminum IDAN frame. The IDAN building block can be used in just about any combination with other IDAN building blocks to create a simple but rugged PC/104 stack. Refer to Appendix C, *IDAN™ Dimensions and Pinout*, for more information. The CME34BT cpuModule can also be purchased as part of a custom-built RTD HiDAN™ or HiDAN*plus* High Reliability Intelligent Data Acquisition Node. Contact RTD for more information on its high reliability PC/104 systems.

### CME34BT Model Options

The basic cpuModule model options are shown below. Refer to the RTD website ([www.rtd.com](http://www.rtd.com)) for more detailed ordering information and any new variations that may be available.

**Table 1 CME34BT cpuModule Model Options<sup>1</sup>**

| Part Number                | Description                                            | Stack Height |
|----------------------------|--------------------------------------------------------|--------------|
| CME34BTS1460HR-4096/S32GX  | Intel Atom® E3815, Single Core 1.46 GHz 4GB DDR3-SDRAM | 15.24mm      |
| CME34BTD1330HR-4096/S32GX  | Intel Atom® E3825, Dual Core 1.33 GHz 4GB DDR3-SDRAM   | 15.24mm      |
| CME34BTQ1910HR-4096/S32GX  | Intel Atom® E3845, Quad Core 1.91 GHz 4GB DDR3-SDRAM   | 22mm         |
| CME34BTS1460HR-4096/S32GXF | CME34BTS1460HR-4096/S32GX with Flat Heatspreader       | 15.24mm      |
| CME34BTD1330HR-4096/S32GXF | CME34BTD1330HR-4096/S32GX with Flat Heatspreader       | 15.24mm      |
| CME34BTQ1910HR-4096/S32GXF | CME34BTQ1910HR-4096/S32GX with Flat Heatspreader       | 22mm         |

1. All ordering options include a Trusted Platform Module (TPM).

### Cable Kits and Accessories

For maximum flexibility, RTD does not provide cables with the cpuModule. You may wish to purchase the CME34BT cpuModule cable kit (P/N XK-CM108), which contains:

- Utility Port 2.0 multi-function cable (2x USB 2.0 ports, battery, reset button, power button, speaker)
- Two serial port cables (DIL-10 to DSUB-9)
- VGA monitor cable (DIL-10 to high density 15-pin DSUB)
- aDIO cable (DIL-16 to DSUB-25)
- Two USB cables (5-pin SIL to USB A)
- PCIe/104 Type 2 break-out board (provides standard PC interfaces for SATA and USB devices)
- Ethernet cable (DIL-10 to RJ-45)

For additional accessories, refer to the RTD website.

# Board Features

- Intel Atom® E3800 Series Processor:

| Part Number  | Cores | Core Frequency | L1 Cache (data)       | L1 Cache (instruction) | L2 Cache               | Stack Height     |
|--------------|-------|----------------|-----------------------|------------------------|------------------------|------------------|
| CME34BTS1460 | One   | 1.46 GHz       | 32 KByte              | 32 KByte               | 512 KByte              | 0.600" (15.24mm) |
| CME34BTD1330 | Two   | 1.33 GHz       | 32 KByte <sup>1</sup> | 32 KByte <sup>1</sup>  | 512 KByte <sup>1</sup> | 0.600" (15.24mm) |
| CME34BTQ1910 | Four  | 1.91 GHz       | 32 KByte <sup>1</sup> | 32 KByte <sup>1</sup>  | 512 KByte <sup>1</sup> | 0.866" (22mm)    |

1. Discreet L1 and L2 cache structures for each core

- Intel Atom® technology
  - E3815 Single Core Processor
  - E3825 Dual Core Processor
  - E3845 Quad Core Processor
- ACPI 5.0 compliant
- Thermal Throttling reduces clock speed to prevent thermal runaway
- 22 nm process
- Max Core Temperature of 110 C
- 2-4 GBytes BGA DDR3 SDRAM with ECC
  - Single-channel memory interface
  - Transfer rates up to 1333 MT/s Data Rate
  - Surface Mounted for maximum reliability
- Stackable 156-pin PCIe/104 Type 2 bus on top
  - One SATA 2.0, Rev 2.0 3GB/s
  - Two USB 2.0
  - ATX Power Supply Signaling
  - RTC Battery
- Stackable 156-pin PCIe/104 Type 2 bus on bottom
  - Four PCI Express x1 Links. Gen 2, 5 GT/s
  - Two USB 2.0
  - ATX Power Supply Signaling
  - RTC Battery
- Advanced Thermal Management
  - Thermal Monitor throttles processor and memory to prevent thermal runaway
  - Passive Fanless Heatsink
  - Modified heatsinks and flat heat spreaders are available
  - Passive Structural Heatsink & Heatpipes in IDAN and HiDAN System Configurations
- Advanced Configuration and Power Interface (ACPI)
  - ACPI 5.0 Compliant
  - Wake events include:
    - USB event (device insertion, keyboard keystroke, etc.)

- Power Switch
  - etc.
- Real-Time Clock (external battery required to maintain time only)
- Nonvolatile storage of CMOS settings without battery
- Advanced Watchdog timer
- Complete PC-compatible Single Board Computer
- Trusted Platform Module (TPM)
  - Fully compliant to the Trusted Computing Group (TCG) Trusted Platform Module (TPM) version 2.0 specification
  - Provides for secure storage of cryptographic keys

## I/O

- Analog VGA Interface
  - RGB CRT output
  - Maximum Resolution 1920 x 1200
- DisplayPort Interface
  - Embedded DisplayPort (eDP) 1.3
  - DisplayPort, DVI, and HDMI
  - Audio over DisplayPort
- Two Gigabit Ethernet Ports
  - Intel WGI210IT PCI Express x1 Controller
  - 10/100/1000 Auto-negotiation
  - Jumbo Frame Support (9.5 kB)
  - Automatic MDI/MDI-X crossover capable
- Software-configurable RS-232/422/485 serial ports
  - 16550 compatible UARTs for high-speed
  - 120 Ohm Termination resistors for RS-422/485 through BIOS Configuration
  - Fully jumperless configuration
  - Increased FIFO size of 512 bytes
  - Supports echo cancellation, Auto-RTS and Auto-CTS
  - Each serial port can be configured as two limited serial ports for a total of four serial ports
- Advanced Digital I/O (aDIO)
  - One 8-bit bit-programmable I/O with Advanced Digital Interrupt Modes
  - One 4-bit port programmable as input or output
  - Event Mode Interrupt generates an interrupt when any input bit changes
  - Match Mode Interrupt generates an interrupt when input bits match a preset value
  - External Strobe Mode latches 8 data inputs and generates an interrupt
  - Two Strobes can be configured as readable inputs
- Nine USB 2.0 (Universal Serial Bus) Ports
  - Supports 480 Mb/s (high-speed), 12Mb/s (full-speed), and 1.5Mbps (low speed) peripherals
  - Automatic resettable over-current protection (500 mA @ 5 Vdc per port)
  - USB Boot capability
  - Five ports on 0.1" I/O headers: two on C17, two on CN27, and one on CN5
  - Four ports on PCIe bus connectors
- Serial ATA (SATA)
  - Two SATA revision 2.0 links
    - One link to the top-side PCIe connector
    - One link to the onboard surface-mount SATA Flash drive
  - Onboard 32 GB SATA Flash drive
    - Built in Wear Leveling, Error Correction and Bad Block Management
    - SMART supported (Self-Monitoring, Analysis and Reporting Technology)
  - Compatibility mode supports legacy operating systems.

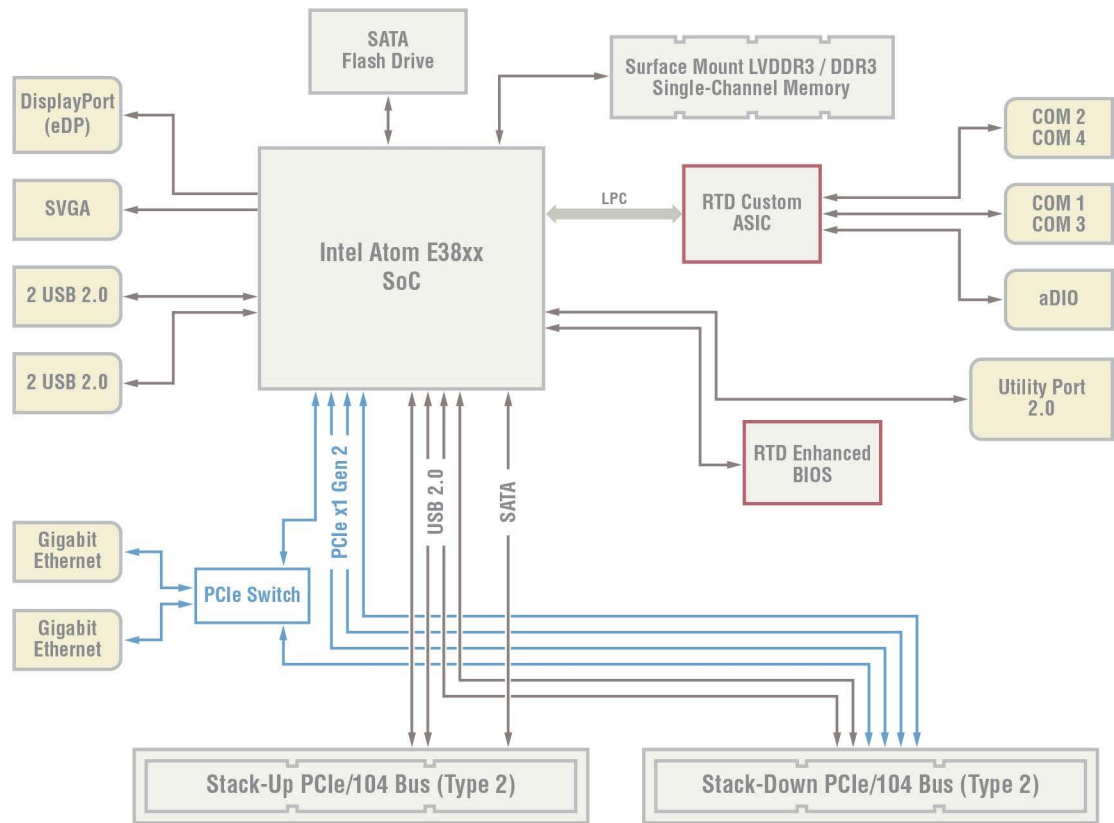
- Utility Port 2.0 Connector
  - 1x USB 2.0 (Universal Serial Bus) Port with over-current protection
  - Speaker port (0.1 W output)
  - Hardware Reset input
  - Soft Power Button input
- Battery input for Real Time Clock
- Power Management
  - ACPI 5.0 Support - Advanced Configuration and Power interface
  - ATX support for “Soft Off”
  - ATX Power signals

## **RTD Enhanced BIOS**

- User-configurable using built-in Setup program
- Flash-based CMOS Setup; no battery required to store CMOS settings
- Supports boot from SATA or USB
- UEFI (Unified Extensible Firmware Interface) Shell
- Special RTD Reliability Enhancements

## Block Diagram

The next figure shows a simplified block diagram of the CME34BT cpuModule.



**Figure 2 CME34BT cpuModule Simplified Block Diagram**

# Specifications

## Physical Characteristics

Basic dimensions and the weight of the CME34BT are listed below. For a more detailed dimensional drawing, refer to the *Physical Dimensions* section on page 78.

- Dimensions
  - Board Size
    - Length (L): 95.885 mm L (3.775")
    - Width (W): 90.170 mm W (3.550")
  - Stand-off Height Above the CPU:
    - Single- and Dual-core models: 15.240 mm (0.866")
    - Quad-core models 22.000 mm (0.866")
  - Stand-off Height Below the CPU:
    - All versions: 15.240mm (0.600")
- Approximate Weight with Heatsink
  - Single-core models: 0.124 kg (0.274 lb)
  - Dual-core models: 0.16 kg (0.354 lb)
  - Quad-core models: 0.184 kg (0.406 lb)
- Approximate IDAN weight
  - All models: 0.63 kg (1.38 lb)

## Power Consumption

Exact power consumption depends on the actual application. Typical power consumption of the CME34BT is listed in Table 3. It is expected that power consumption of a typical application will be a combination of these scenarios.

**Table 2 Typical Power Consumption**

| Part Number         | Typical Power Consumption |
|---------------------|---------------------------|
| CME34BTS615HR-4096  | 6.8 W                     |
| CME34BTD1330HR-4096 | 7.0 W                     |
| CME34BTQ1910HR-4096 | 8.5 W                     |



**Note** The measurements listed above are intended to represent common usage scenarios. The *cpuModule*'s power consumption is heavily influenced by the running software and system activity.

## Operating Conditions

**Table 3 Operating Conditions**

| Symbol                    | Parameter                                  | Test Condition            | Min.    | Max.  |
|---------------------------|--------------------------------------------|---------------------------|---------|-------|
| <b>V<sub>CC5</sub></b>    | 5V Supply Voltage                          |                           | 4.75V   | 5.25V |
| <b>V<sub>CCSTBY</sub></b> | 5V Standby Voltage <sup>1</sup>            |                           | 4.75V   | 5.25V |
| <b>I<sub>CCSTBY</sub></b> | 5V Standby Current <sup>1</sup>            |                           | -       | 500mA |
| <b>T<sub>a</sub></b>      | Ambient Operating Temperature <sup>2</sup> |                           | -40     | +85C  |
| <b>T<sub>s</sub></b>      | Storage Temperature                        |                           | -40     | +70C  |
| <b>R<sub>h</sub></b>      | Humidity                                   | Non-Condensing            | 0       | 90%   |
| <b>MTBF</b>               | Mean Time Before Failure (hours)           | CME34BTS1460HR-4096/S32GX | 354,610 |       |
|                           |                                            | CME34BTD1330HR-4096/S32GX | 354,610 |       |
|                           |                                            | CME34BTQ1910HR-4096/S32GX | 354,610 |       |

1. With supplied heat sink solution. Depending on the CPU usage, performance may degrade as the ambient temperature approaches the maximum. Contact RTD Tech Support for more information.
2. Calculation Model: Telcordia Issue 2; Calculation Method: Method 1 Case 3; Temperature: +30C; Environment: Ground Benign, Controlled

## Electrical Characteristics

The table below lists the Electrical Characteristics of the CME34BT. Operating outside of these parameters may cause permanent damage to the cpuModule.

**Table 4 Electrical Characteristics**

| Symbo<br>l                       | Parameter                             | Test<br>Condition                 | Min.    | Max.   |
|----------------------------------|---------------------------------------|-----------------------------------|---------|--------|
| <b>USB Ports</b>                 |                                       |                                   |         |        |
| <b>I<sub>oc</sub></b>            | Overcurrent Limit                     | Each port                         | 0.5 A   | 5.0 A  |
| <b>SVGA Port</b>                 |                                       |                                   |         |        |
| <b>V<sub>OH</sub></b>            | Output Voltage High<br>HSYNC, VSYNC   | I <sub>OH</sub> = -8.0 mA         | 2.4 V   | 3.3 V  |
| <b>V<sub>OL</sub></b>            | Output Voltage Low<br>HSYNC, VSYNC    | I <sub>OL</sub> = 8.0 mA          | 0.0 V   | 0.5 V  |
| <b>V<sub>OH</sub></b>            | Output Voltage High<br>DDC_*          | I <sub>OH</sub> = -4.0 mA         | 2.4 V   | 3.3 V  |
| <b>V<sub>OL</sub></b>            | Output Voltage Low<br>DDC_*           | I <sub>OL</sub> = 8.0 mA          | 0.0 V   | 0.4 V  |
| <b>V<sub>IH</sub></b>            | Input Voltage High<br>DDC_*           | —                                 | 2.0 V   | 5.5 V  |
| <b>V<sub>IL</sub></b>            | Input Voltage Low<br>DDC_*            | —                                 | -0.3 V  | 0.8 V  |
| <b>I<sub>DDCvcc</sub></b>        | Supply Current for<br>DDC Electronics | —                                 |         | 500 mA |
| <b>DisplayPort</b>               |                                       |                                   |         |        |
| <b>I<sub>VCC</sub></b>           | Supply Current                        | —                                 |         | 500 mA |
| <b>Serial Ports - RS-232</b>     |                                       |                                   |         |        |
| <b>V<sub>OH</sub></b>            | Output Voltage High                   | R <sub>L</sub> = 3 k              | 5.0 V   | 7.0 V  |
| <b>V<sub>OL</sub></b>            | Output Voltage Low                    | R <sub>L</sub> = 3 k              | -7.0 V  | -5.0 V |
| <b>V<sub>IH</sub></b>            | Input Voltage High                    | —                                 | 2.0 V   | 15 V   |
| <b>V<sub>IL</sub></b>            | Input Voltage Low                     | —                                 | -15 V   | 0.6 V  |
| <b>Serial Ports - RS-422/485</b> |                                       |                                   |         |        |
| <b>V<sub>OD1</sub></b>           | Differential Output                   | R <sub>L</sub> = 100 Ohm          | 2.0 V   |        |
| <b>V<sub>OD2</sub></b>           | Differential Output                   | R <sub>L</sub> = 54 Ohm           | 1.5 V   |        |
| <b>V<sub>OC</sub></b>            | Common Mode<br>Output                 | R <sub>L</sub> = 54 or 100<br>Ohm |         | 3.0 V  |
| <b>V<sub>TH</sub></b>            | Differential Input<br>Threshold       |                                   | -200 mV | -50 mV |
| <b>V<sub>I</sub></b>             | Absolute Max Input<br>Voltage         |                                   | -18 V   | 18 V   |

**Table 4 Electrical Characteristics**

| Symbol                                  | Parameter                      | Test Condition           | Min.   | Max.              |
|-----------------------------------------|--------------------------------|--------------------------|--------|-------------------|
| <b>aDIO</b>                             |                                |                          |        |                   |
| $V_{OH}$                                | Output Voltage High            | $I_{OH} = -29\text{ mA}$ | 2.4 V  | 3.3 V             |
| $V_{OL}$                                | Output Voltage Low             | $I_{OL} = 17\text{ mA}$  | 0.0 V  | 0.5 V             |
| $V_{IH}$                                | Input Voltage High             | —                        | 1.8 V  | 5.5 V             |
| $V_{IL}$                                | Input Voltage Low              | —                        | -0.3 V | 0.8 V             |
| $I_{ADIOvc}$                            | Supply current                 | —                        |        | 500 mA            |
| $T_{MIN}$                               | Update Interval                | —                        | 1ms    |                   |
| <b>Utility Port 2.0 Connector (CN5)</b> |                                |                          |        |                   |
| $V_{RTC}$                               | Input RTC Voltage <sup>1</sup> | —                        | 2.3 V  | 3.6 V             |
| $I_{RTC}$                               | RTC Battery Current            | —                        |        | < 4 $\mu\text{A}$ |
| $I_{UTILvc}$                            | Utility Supply Current         | —                        |        | 500 mA            |

1. Only required to maintain date and time when power is completely removed from the system. Not required for board operation.

# Migrating to RTD's Intel E3800 Series cpuModules

When migrating to RTD's Intel E3800 Series cpuModules, there are several differences from some preceding RTD cpuModule product families of which the customer should be aware.

## Connector Pinout & Function Differences

The Intel E3800 Series cpuModules have several connector-related differences, which are summarized below. Complete information about the connectors on the CME34BT can be found in Chapter 3, *Connecting the cpuModule*.

### New I/O Connectors

The CME34BT cpuModule introduces a new I/O connector that was not present on some previous RTD cpuModules:

- **DisplayPort (CN19)**
  - Provides video and audio output
  - Unlike legacy LVDS and flat panel connectors which needed to have their BIOS pre-configured to a specific manufacturer's flat panel parameters, the DisplayPort connector is compatible with any display that has a DisplayPort interface.

### Connector Differences

While some connectors on the CME34BT have identical pinouts as on previous cpuModule generations, the features of the connector are sometimes different. Other connectors have entirely different pinouts than those on previous cpuModule generations. Connectors with pinout and feature differences include:

- **Auxiliary Power Connector (CN3)**
  - 12 pins (unlike the 10-pin connector on RTD Montevina cpuModules)
  - +5V and +12V inputs (unlike +5V only auxiliary power connectors found on previous RTD cpuModule generations)
    - While the RTD Intel E3800 Series CPU only requires +5 volts, +12V may optionally be connected to CN3 if it is required by any other device in the system.
  - The Auxilliary Power Connector excludes ATX power signaling. The ATX power signals are provided on the PCIe/104 Type 2 connector.
- **Utility Port 2.0 (CN5)**
  - Replaces the legacy Utility Port multi-function connector
  - Replaces the PS/2 Mouse and Keyboard connections with one USB 2.0 port
  - The pinout of the Utility Port 2.0 is **not** compatible with previous generations of RTD cpuModules.
- **COM Ports (CN7 and CN8)**
  - Connector **CN7** can be configured to output Serial POST codes out of the TXD pin when in single RS-232 mode (unlike RTD Montevina cpuModules, where the DTR pin was configured for serial POST code output on the second port of CN7 in dual-port mode). For more information, refer to *Serial Power-On-Self-Test (POST) Code Output* on page 72.
  - Of all single and dual-port modes, only the single RS-232 pinout is compatible with previous generations of RTD cpuModules.
  - Dual-mode COM port pinouts permit dual RS-232, dual RS-422, and dual RS-485 modes. Unlike some previous generations of RTD cpuModules, a dual "combined" mode with RS-232 and RS-422/458 is not supported.

- Unlike some previous cpuModule generations, the single-mode RS-422/485 pinouts of the CME34BT do not include the DCD, DSR, DTR, and RI signals.
- When a COM port connector is configured for dual-port mode, enabling/disabling COM port termination enables/disables the termination for both ports on that connector. (Some previous generations of cpuModules have the ability to independently enable/disable termination on both ports when configured for dual-mode.)

## Software Differences

The Intel E3800 Series cpuModules have several software-related differences, which are summarized below. Complete information about these CME34BT features can be found in Chapter 4, *Using the cpuModule*.

### aDIO

While the 16-pin aDIO connector is pin-for-pin compatible with previous generations of RTD cpuModules (such as the Intel Core 2 Duo series), the hardware is slightly different, and requires that writes and reads to/from the aDIO ports happen no more than once per millisecond. Migrating legacy software to the E3800 Series that utilizes RTD's Advanced Digital I/O may require that you add this delay prior to all reads and writes.

### UEFI Shell

Previous generations of RTD cpuModules contained a Failsafe Boot ROM image, which permitted the cpuModule to boot to a DOS prompt even when no disk drives were connected to the system. The RTD E3800 Series cpuModules provide a similar prompt, called the Unified Extensible Firmware Interface (UEFI) Shell, which provides an environment to execute simple commands when no bootable devices are connected to the system.

## Contact Information

RTD Embedded Technologies, Inc.  
103 Innovation Blvd.  
State College, PA 16803-0906  
USA

**Phone:** +1-814-234-8087

**Fax:** +1-814-234-5218

**E-mail:** sales@rtd.com  
techsupport@rtd.com

**Internet:** <http://www.rtd.com>

## Chapter 2 Getting Started

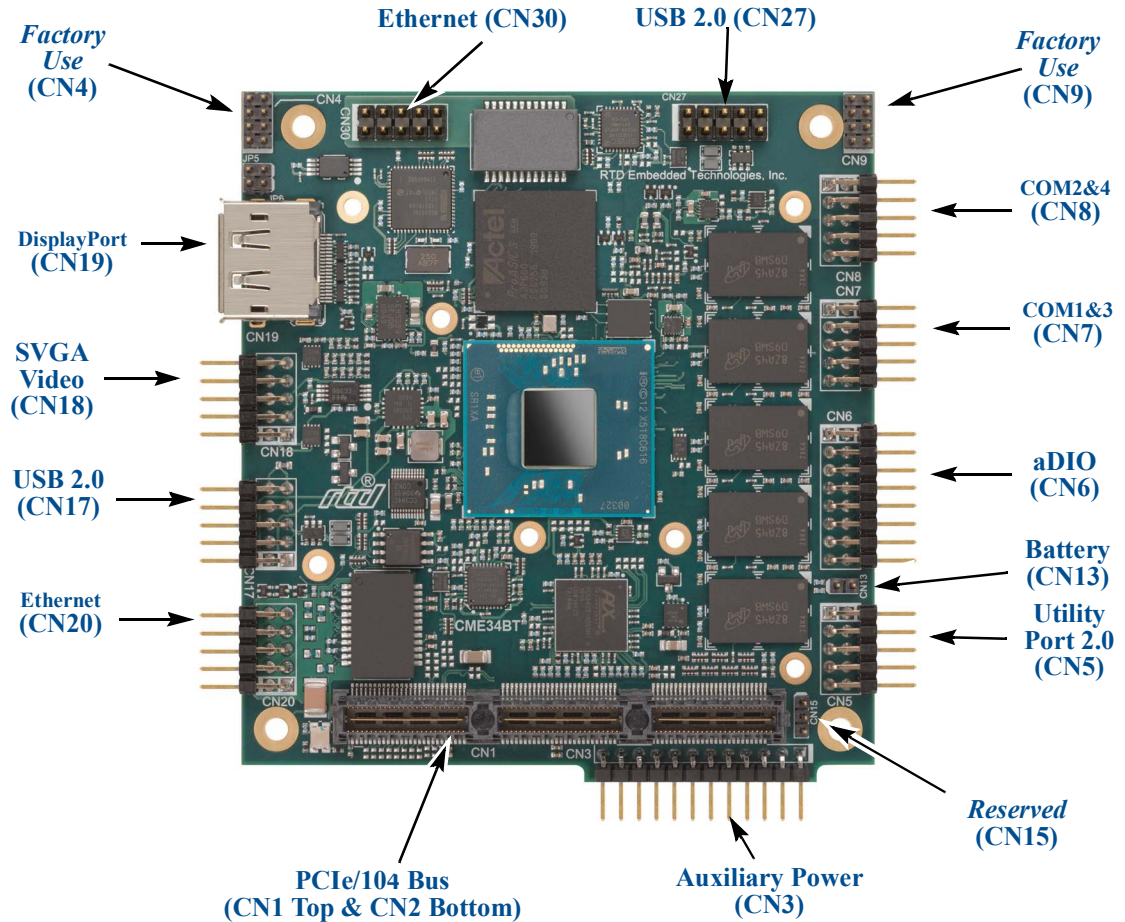
For many users, the factory configuration of the CME34BT cpuModule can be used to get a PC/104 system operational. You can get your system up and running quickly by following the simple steps described in this chapter, which are:

1. Before connecting the cpuModule, the user must be properly grounded to prevent electrostatic discharge (ESD). For more information, refer to *Proper Grounding Techniques* on page 28.
2. Connect power.
3. Connect the Utility Port 2.0 cable.
4. Connect a USB keyboard.
5. Connect a monitor to the SVGA connector or DisplayPort interface.
6. Default BIOS configuration.
7. UEFI (Unified Extensible Firmware Interface) shell.

Refer to the remainder of this chapter for details on each of these steps.

## Connector Locations

Figure 3 shows the connectors of the CME34BT cpuModule.



**Figure 3 CME34BT Connector Locations**



**Note** Pin 1 of each connector is indicated by a white silk-screened square on the top side of the board and a square solder pad on the bottom side of the board. Pin 1 of the bus connectors match when stacking PCIe/104 or PCI-104 modules.

**Table 5 CME34BT Basic Connectors**

| Connector            | Function                     | Size and Pitch                 | Mating Connector                            |
|----------------------|------------------------------|--------------------------------|---------------------------------------------|
| <b>CN1</b>           | PCIe/104 Type 2 Bus (Top)    | 156-pin,<br>0.635mm            | Samtec ASP-129646-03                        |
| <b>CN2</b>           | PCIe/104 Type 2 Bus (Bottom) | 156-pin,<br>0.635mm            | Samtec ASP-129637-03                        |
| <b>CN3</b>           | Auxiliary Power              | 1x12, 0.1"                     | FCI 65039-025LF (housing model number only) |
| <b>CN5</b>           | Utility Port 2.0             | 2x5, 0.1"                      | 3M 89110-0001                               |
| <b>CN6</b>           | aDIO                         | 2x8, 0.1"                      | 3M 89116-0001                               |
| <b>CN7</b>           | Serial Port 1 (COM 1&3)      | 2x5, 0.1"                      | 3M 89110-0001                               |
| <b>CN8</b>           | Serial Port 2 (COM 2&4)      | 2x5, 0.1"                      | 3M 89110-0001                               |
| <b>CN13</b>          | RTC Battery Input (optional) | 1x2, 2mm                       | FCI 69305-002LF                             |
| <b>CN15</b>          | <i>Reserved</i>              | 1x3, 2mm                       | FCI 69305-003LF                             |
| <b>CN17 and CN27</b> | USB 2.0                      | 2x5, 0.1"                      | 3M 89110-0001                               |
| <b>CN18</b>          | Video (SVGA)                 | 2x5, 0.1"                      | 3M 89110-0001                               |
| <b>CN19</b>          | DisplayPort                  | Molex P/N Series<br>47272-xxxx | Molex P/N Series<br>68783-xxxx              |
| <b>CN20 and CN30</b> | Ethernet                     | 2x5, 0.1"                      | 3M 89110-0001                               |



**WARNING** If you connect power incorrectly, the module will almost certainly be damaged or destroyed. Such damage is not covered by the RTD warranty! Please verify connections to the module before applying power.

Power is normally supplied to the cpuModule through the top or bottom PCIe connectors (**CN1** or **CN2**). If you are placing the cpuModule onto a stack that has a PCIe/104 power supply, you do not need to make additional connections to supply power.

Alternatively, if you are using the cpuModule without a PC/104 stack or with a stack that does not include a power supply, refer to *Auxiliary Power (CN3)* on page 30 for an alternative method on how to power the cpuModule.

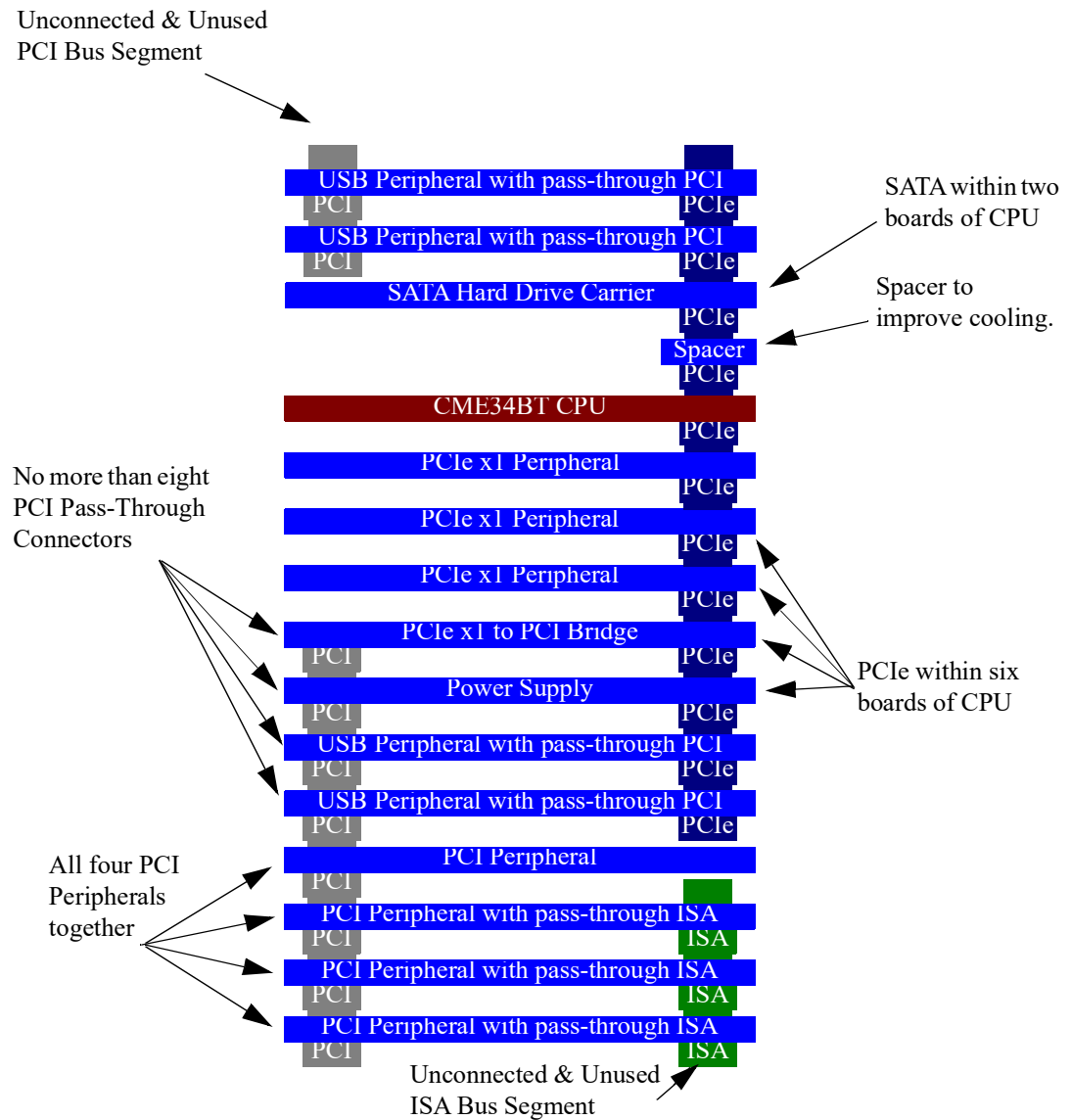
## Selecting the Stack Order for the CME34BT

There are several things to consider when selecting the order of boards in the stack. Before selecting the order, be sure to determine which bus connector on each board is the “Active” bus. Typically, if a peripheral module has both PCIe and PCI bus connectors, only the PCIe is active and the PCI is pass-through. The following is a list of rules to use to determine the stack order:

1. The PCIe connectors above and below the CPU have completely separate signals. Therefore it is possible to attach boards to the PCIe connector above and below the CPU.
2. Any board that uses a SATA link must be within one board of the CPU.
3. Any board that uses a PCIe link must be within six boards of the CPU. Peripheral boards that repopulate a PCIe link should be placed closer to the CPU than any peripheral boards that use a PCIe link without repopulating.
4. To preserve power integrity, it is recommended that there be no more than six boards between the CPU and the power supply.
5. In order to maintain maximum performance over the full temperature range, it is recommended that a PCIe spacer be used between the CPU and any board immediately above it.
6. A maximum of four PCI boards may be attached to any PCI bus. If more than four PCI boards are needed in a system, a PCIe to PCI bridge is required.
7. The CPU or PCIe to PCI bridge (if one exists in the stack) must be at one end of the PCI bus segment, and all of the peripheral cards at the other end. There may be up to eight PCI pass-through connectors between the CPU or PCIe to PCI bridge and the peripheral cards.
8. There must be no more than two boards between the first PCI peripheral and the last PCI peripheral. If there are four PCI peripheral cards in a PCI bus segment, there may not be any PCI pass-through connections between them.
9. The standard CPU has the PCI connection below the board only. This means that all boards with an active PCI bus interface must be below the CPU. Contact RTD sales if it is required to have a PCI board above the CPU.

## Stack Example

The figure below shows an example of a complete system stack. Most systems will be a subset of this example. This example stack may be further expanded with PCIe to PCIe bridges or a PCIe to PCI bridge.



**Figure 4 System Stacking Example**

## Connecting to the Stack

The bus connectors of the cpuModule are simply plugged onto a PC/104 stack to connect to other devices. Follow the procedure below to ensure that stacking of the modules does not damage connectors or electronics.



**WARNING** Do not force the module onto the stack! Wiggling the module or applying too much pressure may damage it. If the module does not readily press into place, remove it, check for bent pins or out-of-place keying pins, and try again.

---

For mechanical dimensions, including board-to-board spacing, see *Physical Dimensions* on page 78.

1. Turn off power to the PC/104 system or stack.
2. Always work at an ESD protected workstation, and wear a grounded wrist-strap.
3. Select and install stand-offs to properly position the cpuModule on the stack.
4. Remove the cpuModule from its anti-static bag.
5. Check that pins of the bus connector are properly positioned.
6. Check the stacking order; make sure all of the busses used by the peripheral cards are connected to the cpuModule.
7. Hold the cpuModule by its edges and orient it so the bus connector pins line up with the matching connector on the stack.
8. Gently and evenly press the cpuModule onto the PC/104 stack.

## Power Input Connections

Power to the board must come from either the top or bottom PCIe/104 Type 2 bus connectors (**CN1** or **CN2**), or the auxiliary power connector (**CN3**). These connectors provide the required +5V DC voltage rail input to the cpuModule. While the RTD Intel E3800 Series CPU only requires +5 volts, +12V may optionally be connected to CN3 if it is required by any other device in the system.

## Connecting to the Utility Port 2.0 Connector

The Utility Port 2.0 connector (**CN5**) implements the following interfaces:

- One USB 2.0 (Universal Serial Bus) port
- Speaker port (0.1 W output)
- Hardware Reset input
- Battery input for Real Time Clock
- Soft Power Button input

To use these interfaces, you must connect to the Utility Port 2.0 connector (**CN5**). The Utility Port 2.0 cable from the RTD cable kit provides a small speaker, two USB 2.0 ports for a keyboard and mouse, a push-button for resetting the system, a soft-power button, and a lithium battery to provide backup power for the real time clock.

Refer to *Utility Port 2.0 Connector (CN5)* on page 32 to connect devices to the Utility Port 2.0 connector.

## Booting the CME34BT cpuModule for the First Time

You can now apply power to the cpuModule. If you press **ESC** on a keyboard while booting, the cpuModule will enter Setup. Once you have configured the cpuModule using Setup, save your changes and reboot. If you don't press **ESC**, the cpuModule will boot using the current settings.



**Note** You may miss the initial sign-on messages if your monitor takes a while to power on.

**Note** By default, cpuModules are shipped with a UEFI (Unified Extensible Firmware Interface) shell. When no other bootable device is installed in the system, the system will boot to it exclusively. This internal bootable shell may be disabled in the BIOS screen's boot menu.

---



**WARNING** Incorrect modifications to the BIOS can cause your system to break. If you are unsure of any changes made to the BIOS there is a **Load RTD Defaults** option under the **Exit** tab in the **SCU**. This option restores the BIOS to the default factory settings.

---



## Chapter 3 Connecting the cpuModule

This chapter provides information on all CME34BT cpuModule connectors.

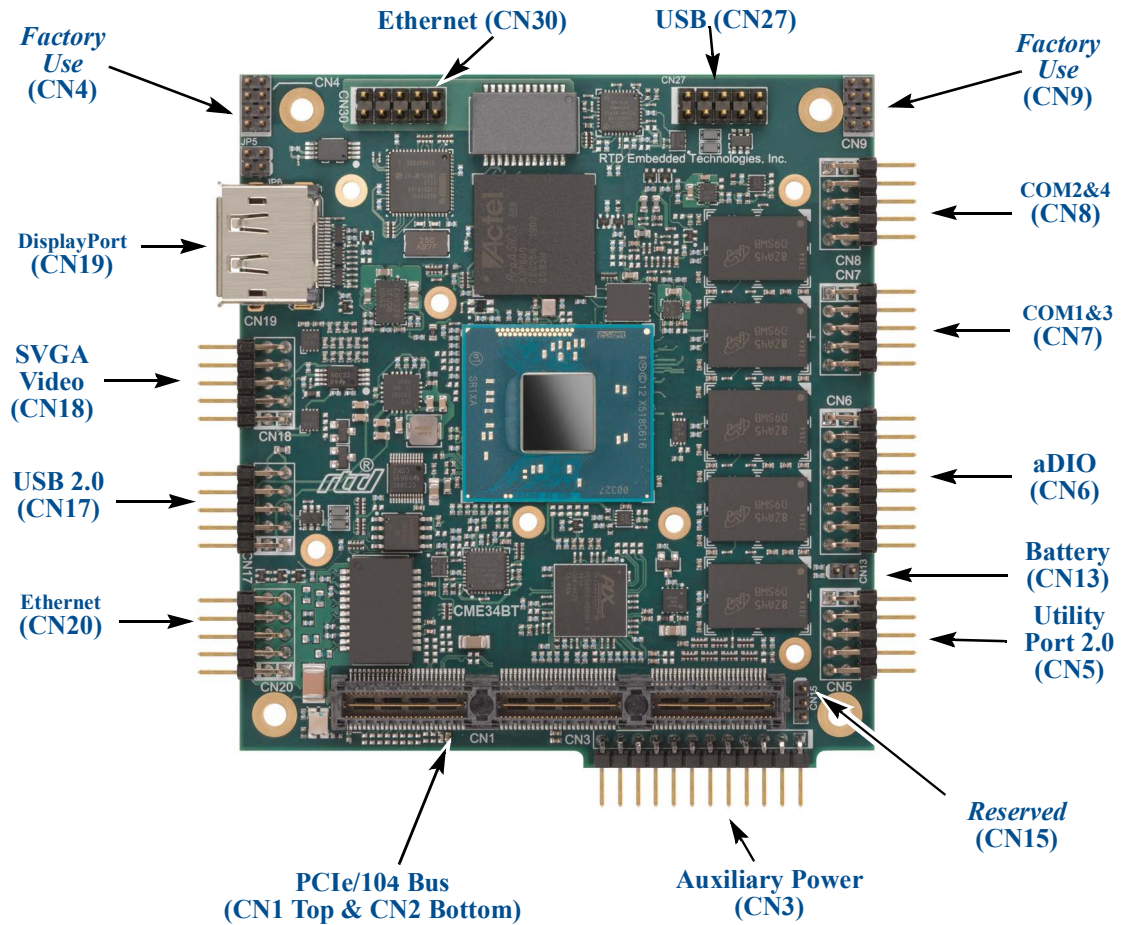
|                                                                    |          |
|--------------------------------------------------------------------|----------|
| <i>Proper Grounding Techniques</i>                                 | —page 28 |
| <i>Connector Locations</i>                                         | —page 28 |
| <i>Auxiliary Power (CN3)</i>                                       | —page 30 |
| <i>Utility Port 2.0 Connector (CN5)</i>                            | —page 32 |
| <i>SVGA Video Connector (CN18)</i>                                 | —page 35 |
| <i>DisplayPort Connector (CN19)</i>                                | —page 36 |
| <i>Serial Port 1 (CN7) and Serial Port 2 (CN8)</i>                 | —page 38 |
| <i>Advanced Digital I/O (aDIO™) Port (CN6)</i>                     | —page 44 |
| <i>USB 2.0 Connector (CN17, CN27)</i>                              | —page 45 |
| <i>Ethernet (10/100/1000Base-T and -TX) Connector (CN20, CN30)</i> | —page 47 |
| <i>PCIe/104 Type 2 Bus (CN1 - Top and CN2 - Bottom)</i>            | —page 48 |
| <i>Optional RTC Battery Input (CN13)</i>                           | —page 52 |
| <i>Reserved (CN15)</i>                                             | —page 52 |

## Proper Grounding Techniques

Before removing the CME34BT from its static bag, proper grounding techniques must be used to prevent electrostatic discharge (ESD) damage to the cpuModule. Common grounding procedures include an anti-static mat on a workbench, which may connect to an anti-static wrist strap (also known as an ESD wrist strap) on the wrist of the technician or engineer.

## Connector Locations

Figure 5 shows the connectors of the CME34BT cpuModule.



**Figure 5 CME34BT cpuModule (top view)**



**Note** Pin 1 of each connector is indicated by a white silk-screened square on the top side of the board and a square solder pad on the bottom side of the board. Pin 1 of the bus connectors match when stacking PC/104 modules.

**Table 6 CME34BT Basic Connectors**

| Connector                | Function                        | Size and Pitch                    | Mating Connector                               |
|--------------------------|---------------------------------|-----------------------------------|------------------------------------------------|
| <b>CN1</b>               | PCIe/104 Type 2 Bus (Top)       | 156-pin,<br>0.635mm               | Samtec ASP-129646-03                           |
| <b>CN2</b>               | PCIe/104 Type 2 Bus<br>(Bottom) | 156-pin,<br>0.635mm               | Samtec ASP-129637-03                           |
| <b>CN3</b>               | Auxiliary Power                 | 1x12, 0.1"                        | FCI 65039-025LF (housing<br>model number only) |
| <b>CN5</b>               | Utility Port 2.0                | 2x5, 0.1"                         | 3M 89110-0001                                  |
| <b>CN6</b>               | aDIO                            | 2x8, 0.1"                         | 3M 89116-0001                                  |
| <b>CN7</b>               | Serial Port 1 (COM1&3)          | 2x5, 0.1"                         | 3M 89110-0001                                  |
| <b>CN8</b>               | Serial Port 2 (COM2&4)          | 2x5, 0.1"                         | 3M 89110-0001                                  |
| <b>CN13</b>              | RTC Battery Input (optional)    | 1x2, 2mm                          | FCI 69305-002LF                                |
| <b>CN15</b>              | <i>Reserved</i>                 | 1x3, 2mm                          | FCI 69305-003LF                                |
| <b>CN17 and<br/>CN27</b> | USB 2.0                         | 2x5, 0.1"                         | 3M 89110-0001                                  |
| <b>CN18</b>              | Video (SVGA)                    | 2x5, 0.1"                         | 3M 89110-0001                                  |
| <b>CN19</b>              | DisplayPort                     | Molex P/N<br>Series<br>47272-xxxx | Molex P/N Series<br>68783-xxxx                 |
| <b>CN20 and<br/>CN30</b> | Ethernet                        | 2x5, 0.1"                         | 3M 89110-0001                                  |

## Auxiliary Power (CN3)

The Auxiliary Power connector (**CN3**) can be used to supply power to devices that are attached to the cpuModule. These devices include hard drive, front-end boards for data acquisition systems, and other devices.

Power can also be conveyed to the module through the Auxiliary Power connector (**CN3**). The Intel E3800 Series cpuModule only requires +5 VDC and ground for operation. +12 V may optionally be connected to CN3 if it is required by any other device in the system.



**Note** Although it is possible to power the cpuModule through the Auxiliary Power connector, the preferred method is to power it through the bus connector from a power supply in the stack. The cpuModule can have large current transients during operation, which make powering it through wires difficult. Powering through the bus eliminates such problems as voltage drop and lead inductance.



**WARNING** System configurations which are powered using CN3 should **only** be used power the CPU. The connector is not intended to power configurations which consist of peripheral cards.

If using the Auxiliary Power connector to power the system, care must be taken to ensure good power connections. The power and ground leads must be twisted together, or as close together as possible to reduce lead inductance. A separate lead must be used for each of the power pins. All 5V pins and all ground pins must be connected. Do not use wire smaller than 20 gauge, and take care to ensure the length of the wire does not exceed 2 ft. The power supply solution must be verified by measuring voltage at the Auxiliary Power Connector and verifying that it meets the input voltage specifications. The voltage at the connector should be checked with an oscilloscope while the system is operational.



**WARNING** This 12-pin power connector is **not** compatible with previous generations of RTD cpuModules.

**WARNING** If you connect power incorrectly, the module will almost certainly be destroyed. Please verify power connections to the module before applying power.

**Table 7 Auxiliary Power Connector (CN3)**

| Pin | Signal             | Function     |
|-----|--------------------|--------------|
| 1   | GND                | Ground       |
| 2   | +5 V               | +5 Volts DC  |
| 3   | +5 V               | +5 Volts DC  |
| 4   | GND                | Ground       |
| 5   | GND                | Ground       |
| 6   | +12 V <sup>1</sup> | +12 Volts DC |
| 7   | +12 V <sup>1</sup> | +12 Volts DC |
| 8   | GND                | Ground       |
| 9   | GND                | Ground       |
| 10  | +5 V               | +5 Volts DC  |
| 11  | +5 V               | +5 Volts DC  |
| 12  | GND                | Ground       |

1. +12 V not required for operation of the cpuModule



**WARNING** *The user is responsible for any over-voltage protection that would be necessary to power through CN3. Over-voltage damage at CN3 is not covered under warranty.*

---

## Utility Port 2.0 Connector (CN5)

The Utility Port 2.0 connector implements the following functions:

- 1x USB 2.0 (Universal Serial Bus) Port
- Speaker port (0.1 W output)
- Hardware Reset input
- Soft Power Button input
- Battery input for Real Time Clock

Table 8 provides the pinout of the Utility Port 2.0 connector.

**Table 8 Utility Port 2.0 Connector (CN5)**

| Pin | Signal     | Function                         | In/Out |
|-----|------------|----------------------------------|--------|
| 1   | SPKR       | Speaker Output (open collector)  | out    |
| 2   | PWR        | +5 V                             | out    |
| 3   | RESET#     | Manual Push-Button Reset         | in     |
| 4   | DATA–      | Bidirectional data line for USB1 | in/out |
| 5   | PWRSW#     | Soft Power Button                | in     |
| 6   | DATA+      | Bidirectional data line for USB1 | in/out |
| 7   | GND        | Ground                           | out    |
| 8   | GND        | Ground                           | out    |
| 9   | BAT        | RTC Battery Input                | in     |
| 10  | Shield GND | Shield Ground                    | out    |



**WARNING** The pinout of the Utility Port 2.0 connector is **not** compatible with previous generations of RTD cpuModules. Attaching a legacy Utility Port harness to the Utility Port 2.0 connector may damage or destroy the cpuModule.

Facing the connector pins, the pinout is:

|            |     |        |        |      |
|------------|-----|--------|--------|------|
| 9          | 7   | 5      | 3      | 1    |
| BAT        | GND | PWRSW# | RESET# | SPKR |
| Shield GND | GND | DATA+  | DATA-  | PWR  |
| 10         | 8   | 6      | 4      | 2    |

## USB 2.0 Connector

One USB 2.0 compliant connector is available on connector **CN5**. Table 8 provides the pinout of the USB connector.



**Note** For proper operation at USB 2.0 speeds, be sure to use a cable that is rated for USB 2.0, such as the cable kit supplied by RTD.

---

## Speaker

A speaker output is available on pins 1 and 2 of the Utility Port 2.0 connector. These outputs are controlled by a transistor to supply 0.1 W of power to an external speaker. The external speaker should have 8 Ohm impedance and be connected between pins 1 and 2.

## System Reset

Pin 3 of the Utility Port 2.0 connector allows connection of an external push-button to manually reset the system. The push-button should be normally open, and connect to ground when pushed. The type of reset generated by this button can be set in the BIOS configuration utility.

## Soft Power Button

Pin 5 of the Utility Port 2.0 connector allows connection of an external push-button to send a soft power signal to the system. The push-button should be normally open, and connect to ground when pushed. For more information on the modes of the Soft Power Button, refer to the *Power Management* section in Chapter 4, *Using the cpuModule*

## Battery

Pin 9 of the Utility Port 2.0 connector is the connection for an external backup battery. This battery is used by the cpuModule when system power is removed in order to preserve the date and time of the real time clock.

Connecting a battery is only required to maintain time when power is completely removed from the cpuModule. A battery is not required for board operation.



**WARNING** The optional RTC battery input connector (**CN13**) should be left unconnected if the multi-function connector (**CN5**) has a battery connected to pin 9.

---

## SVGA Video Connector (CN18)

Table 9 provides the pinout of the video connector.

**Table 9 SVGA Video Connector (CN18)**

| Pin | Signal | Function                     | In/Out        |
|-----|--------|------------------------------|---------------|
| 1   | VSYNC  | Vertical Sync                | out           |
| 2   | HSYNC  | Horizontal Sync              | out           |
| 3   | DDCSCL | Monitor Communications Clock | out           |
| 4   | RED    | Red Analog Output            | out           |
| 5   | DDCSDA | Monitor Communications Data  | bidirectional |
| 6   | GREEN  | Green Analog Output          | out           |
| 7   | PWR    | +5 V                         | out           |
| 8   | BLUE   | Blue Analog Output           | out           |
| 9   | GND    | Ground                       | out           |
| 10  | GND    | Ground                       | out           |

Facing the connector pins of the SVGA Video connector (CN18), the pinout is:

|            |             |               |               |              |
|------------|-------------|---------------|---------------|--------------|
| 9          | 7           | 5             | 3             | 1            |
| <b>GND</b> | <b>PWR</b>  | <b>DDCSDA</b> | <b>DDCSCL</b> | <b>VSYNC</b> |
| <b>GND</b> | <b>BLUE</b> | <b>GREEN</b>  | <b>RED</b>    | <b>HSYNC</b> |
| 10         | 8           | 6             | 4             | 2            |

## DisplayPort Connector (CN19)

The DisplayPort connector on the CME34BT cpuModule is a standard PC DisplayPort connector complete with latch holes to provide a rugged connecting solution for latching DisplayPort cables.

The DisplayPort supports all mandatory features of the VESA Embedded DisplayPort (eDP) 1.3 standard and supports audio over the connection. Full bandwidth transmission is supported over a two meter cable, with reduced bandwidth transmission supported up to a length of 15 meters.

The connection can directly output single-link HDMI and DVI signals by using a simple passive adapter. Passive adapters achieve HDMI and DVI signaling by adjusting the lower signal levels output by the connector.

Table 10 provides the pinout of the DisplayPort connector.

**Table 10 DisplayPort Connector (CN19)**

| Pin | Signal | Standard Function            | DVI / HDMI mode <sup>1</sup>              | In/Out |
|-----|--------|------------------------------|-------------------------------------------|--------|
| 1   | LN0+   | Main Link, Lane 0 (positive) | TMDS Channel 2 (positive)                 | out    |
| 2   | GND    | Ground                       | Ground                                    | out    |
| 3   | LN0-   | Main Link, Lane 0 (negative) | TMDS Channel 2 (negative)                 | out    |
| 4   | LN1+   | Main Link, Lane 1 (positive) | TMDS Channel 1 (positive)                 | out    |
| 5   | GND    | Ground                       | Ground                                    | out    |
| 6   | LN1-   | Main Link, Lane 1 (negative) | TMDS Channel 1 (negative)                 | out    |
| 7   | LN2+   | Main Link, Lane 2 (positive) | TMDS Channel 0 (positive)                 | out    |
| 8   | GND    | Ground                       | Ground                                    | out    |
| 9   | LN2-   | Main Link, Lane 2 (negative) | TMDS Channel 0 (negative)                 | out    |
| 10  | LN3+   | Main Link, Lane 3 (positive) | TMDS Clock (positive)                     | out    |
| 11  | GND    | Ground                       | Ground                                    | out    |
| 12  | LN3-   | Main Link, Lane 3 (negative) | TMDS Clock (negative)                     | out    |
| 13  | CFG1   | Configuration Pin 1          | Cable Adapter Detect                      | out    |
| 14  | CFG2   | Configuration Pin 2          | Consumer Electronics Control <sup>2</sup> | out    |
| 15  | AUX+   | Auxiliary Channel (positive) | DDC Clock                                 | in/out |
| 16  | GND    | Ground                       | Ground                                    | out    |
| 17  | AUX-   | Auxiliary Channel (negative) | DDC Data                                  | in/out |
| 18  | HPD    | Hot Plug Detect              | Hot Plug Detect                           | in     |
| 19  | DPG    | Return for DPV               | Return for DPV                            | out    |
| 20  | DPV    | +3.3V DC Power               | +3.3V DC Power                            | out    |

1. Requires special passive adapter

2. HDMI mode only

Facing the connector pins of the DisplayPort connector (CN19), the pinout is:

|     |      |      |      |      |      |      |      |      |      |
|-----|------|------|------|------|------|------|------|------|------|
| 19  | 17   | 15   | 13   | 11   | 9    | 7    | 5    | 3    | 1    |
| DPG | AUX- | AUX+ | CFG1 | GND  | LN2- | LN2+ | GND  | LN0- | LN0+ |
| DPV | HPD  | GND  | CFG2 | LN3- | LN3+ | GND  | LN1- | LN1+ | GND  |
| 20  | 18   | 16   | 14   | 12   | 10   | 8    | 6    | 4    | 2    |

## Serial Port 1 (CN7) and Serial Port 2 (CN8)

Serial Port 1 (COM1) is implemented on connector CN7, and Serial Port 2 (COM2) is implemented on connector CN8. The serial ports are normally configured as PC compatible full-duplex RS-232 ports, but you may use the BIOS Setup program to reconfigure these ports as half-duplex RS-422 or full-duplex RS-422 or RS-485. If you reconfigure the ports, you must also select the I/O address and corresponding interrupt using Setup. Table 11 provides the standard I/O addresses and corresponding interrupts.

**Table 11 Serial Port Settings**

| I/O Address (hex) | IRQ  |
|-------------------|------|
| 03F8              | IRQ4 |
| 02F8              | IRQ3 |
| 03E8              | IRQ4 |
| 02E8              | IRQ3 |

### Serial Port UART

The serial ports are implemented with a 16550/16750-compatible UART (Universal Asynchronous Receiver/ Transmitter). This UART is capable of baud rates up to 115.2 kbaud, and includes a 512-byte FIFO. Refer to any standard PC-AT hardware reference for the register map of the UART.

It is possible to change the slew rate for the UARTs of the cpuModule to allow the serial ports to operate at higher speeds than 115200 bps. For more information see [Chapter 4 - “Non-Standard Serial Port Modes”](#).

### RS-232 Serial Port (Default)

The default serial port mode is full-duplex RS-232. With this mode enabled, the serial port connectors must be connected to RS-232 compatible devices. Table 12 provides the serial port connector pinout and shows how to connect to an external DB-25 or DB-9 compatible serial connector.

**Table 12 Serial Port in RS-232 Mode**

| Pin | Signal | Function            | In/Out | DB-25 | DB-9 |
|-----|--------|---------------------|--------|-------|------|
| 1   | DCD    | Data Carrier Detect | in     | 8     | 1    |
| 2   | DSR    | Data Set Ready      | in     | 6     | 6    |
| 3   | RXD    | Receive Data        | in     | 3     | 2    |
| 4   | RTS    | Request To Send     | out    | 4     | 7    |
| 5   | TXD    | Transmit Data       | out    | 2     | 3    |
| 6   | CTS    | Clear To Send       | in     | 5     | 8    |
| 7   | DTR    | Data Terminal Ready | out    | 20    | 4    |
| 8   | RI     | Ring Indicate       | in     | 22    | 9    |
| 9   | GND    | Signal Ground       | —      | 7     | 5    |
| 10  | GND    | Signal Ground       | —      | —     | —    |

Facing the serial port's connector pins, the pinout is:

|     |     |     |     |     |
|-----|-----|-----|-----|-----|
| 9   | 7   | 5   | 3   | 1   |
| GND | DTR | TXD | RXD | DCD |
| GND | RI  | CTS | RTS | DSR |
| 10  | 8   | 6   | 4   | 2   |

## RS-422 or RS-485 Serial Port

You may use the BIOS setup utility to configure the serial ports as RS-422 or RS-485. In this case, you must connect the serial port to an RS-422 or RS-485 compatible device.

When using RS-422 mode, you can use the serial ports in either half-duplex (two-wire) or full-duplex (four-wire) configurations.



**Note** The *cpuModule* has a 120 Ohm termination resistor. Termination is usually necessary on all RS-422 receivers and at the ends of the RS-485 bus. Termination resistors can be enabled in the BIOS setup utility.

When using full-duplex in RS-422 mode, connect the ports as shown in Table 13.

**Table 13 Full-Duplex Connections**

| Port 1 | Port 2 |
|--------|--------|
| RXD+   | TXD+   |
| TXD+   | RXD+   |
| RXD–   | TXD–   |
| TXD–   | RXD–   |

In RS-485 mode, the connection of the ports is always half-duplex, as the transceivers' transmitters are connected to the receivers internally.

## RS-422 Mode Pinout

Table 14 provides the serial port connector pinout when RS-422 mode is enabled.

**Table 14 Serial Port in RS-422 Mode**

| Pin | Signal | Function          | In/Out | DB-9 |
|-----|--------|-------------------|--------|------|
| 1   | TXD-   | Transmit Data (-) | out    | 1    |
| 2   | —      | Reserved          | —      | 6    |
| 3   | TXD+   | Transmit Data (+) | out    | 2    |
| 4   | —      | Reserved          | —      | 7    |
| 5   | RXD+   | Receive Data (+)  | in     | 3    |
| 6   | —      | Reserved          | —      | 8    |
| 7   | RXD-   | Receive Data (-)  | in     | 4    |
| 8   | —      | Reserved          | —      | 9    |
| 9   | GND    | Signal Ground     | —      | 5    |
| 10  | GND    | Signal Ground     | —      | —    |

Facing the serial port connector, the pinout is:

|     |      |      |      |      |
|-----|------|------|------|------|
| 9   | 7    | 5    | 3    | 1    |
| GND | RXD- | RXD+ | TXD+ | TXD- |
| GND | Rsvd | Rsvd | Rsvd | Rsvd |
| 10  | 8    | 6    | 4    | 2    |



**WARNING** The pinout of the COM ports in RS-422 mode is **not** compatible with previous generations of RTD cpuModules.

## RS-485 Mode Pinout

Table 15 provides the serial port connector pinout when RS-485 mode is enabled

**Table 15 Serial Port in RS-485 Mode**

| Pin | Signal | Function      | In/Out | DB-9 |
|-----|--------|---------------|--------|------|
| 1   | D-     | Data (-)      | in/out | 1    |
| 2   | —      | Reserved      | —      | 6    |
| 3   | D+     | Data (+)      | in/out | 2    |
| 4   | —      | Reserved      | —      | 7    |
| 5   | —      | Reserved      | —      | 3    |
| 6   | —      | Reserved      | —      | 8    |
| 7   | —      | Reserved      | —      | 4    |
| 8   | —      | Reserved      | —      | 9    |
| 9   | GND    | Signal Ground | —      | 5    |
| 10  | GND    | Signal Ground | —      | —    |

Facing the serial port connector, the pinout is:

|     |      |      |      |      |
|-----|------|------|------|------|
| 9   | 7    | 5    | 3    | 1    |
| GND | Rsvd | Rsvd | D+   | D-   |
| GND | Rsvd | Rsvd | Rsvd | Rsvd |
| 10  | 8    | 6    | 4    | 2    |



**WARNING** The pinout of the COM ports in RS-485 mode is **not** compatible with previous generations of RTD cpuModules.



**Note** When using the serial port in RS-485 mode, the serial transmitters are enabled and disabled under software control. The transmitters are enabled by manipulating the Request To Send (RTS\*) signal of the serial port controller. This signal is controlled by writing bit 1 of the Modem Control Register (MCR) as follows:

- If MCR bit 1 = 1, then RTS\* = 0, and serial transmitters are disabled
- If MCR bit 1 = 0, then RTS\* = 1, and serial transmitters are enabled

**Note** For more information on the serial port registers, including the MCR, refer to the Serial Port Programming reference in Appendix D.

## Dual Serial Port Modes

The serial port connectors can be configured as dual serial ports in the BIOS. This enhancement in dual RS-232 comes at the cost of losing the DSR and DTR signals, which are only available in single RS-232 mode.

The mapping between the connectors and COM port numbers is shown in Table 16. The supported combinations of serial port modes are listed in Table 17, which also includes a reference to the corresponding connector pinout.

**Table 16 Dual Serial Port Connections**

| Connector | COM A | COM B |
|-----------|-------|-------|
| CN7       | COM 1 | COM 3 |
| CN8       | COM 2 | COM 4 |

**Table 17 Dual Serial Port Modes**

| COM A  | COM B  | Pinout Reference | Echo Mode              |
|--------|--------|------------------|------------------------|
| RS-232 | RS-232 | Table 18         | —                      |
| RS-422 | RS-422 | Table 19         | —                      |
| RS-485 | RS-485 | Table 20         | echo canceled          |
| RS-485 | RS-485 | Table 19         | with echo <sup>1</sup> |

1. Dual RS-485 mode with echo uses dual RS-422 mode pinout



**WARNING** The dual serial port mode pinouts of the COM ports are **not** compatible with previous generations of RTD cpuModules.



**Note** The cpuModule has 120 Ohm termination resistors. Termination is usually necessary on all RS-422 receivers and at the ends of the RS-485 bus. Termination resistors can be enabled in the BIOS setup utility. When termination is enabled in dual port mode, it is enabled for both ports.

**Table 18 COM A (RS-232) and COM B (RS-232)**

| Pin | Signal | Function                    | In/Out | DB-9 |
|-----|--------|-----------------------------|--------|------|
| 1   | DCD1   | COM A - Data Carrier Detect | in     | 1    |
| 2   | CTS1   | COM A - Clear To Send       | in     | 6    |
| 3   | RXD1   | COM A - Receive Data        | in     | 2    |
| 4   | TXD2   | COM B - Transmit Data       | out    | 7    |
| 5   | TXD1   | COM A - Transmit Data       | out    | 3    |
| 6   | RXD2   | COM B - Receive Data        | in     | 8    |
| 7   | RTS1   | COM A - Request To Send     | out    | 4    |

**Table 18 COM A (RS-232) and COM B (RS-232)**

| Pin | Signal | Function              | In/Out | DB-9 |
|-----|--------|-----------------------|--------|------|
| 8   | RI1    | COM A - Ring Indicate | in     | 9    |
| 9   | GND    | Signal Ground         | —      | 5    |
| 10  | GND    | Signal Ground         | —      | —    |

**Table 19 COM A (RS-422) and COM B (RS-422)<sup>1</sup>**

| Pin | Signal | Function                  | In/Out | DB-9 |
|-----|--------|---------------------------|--------|------|
| 1   | TXD1-  | COM A - Transmit Data (–) | out    | 1    |
| 2   | TXD2-  | COM B - Transmit Data (–) | out    | 6    |
| 3   | TXD1+  | COM A - Transmit Data (+) | out    | 2    |
| 4   | TXD2+  | COM B - Transmit Data (+) | out    | 7    |
| 5   | RXD1+  | COM A - Receive Data (+)  | in     | 3    |
| 6   | RXD2+  | COM B - Receive Data (+)  | in     | 8    |
| 7   | RXD1-  | COM A - Receive Data (–)  | in     | 4    |
| 8   | RXD2-  | COM B - Receive Data (–)  | in     | 9    |
| 9   | GND    | Signal Ground             | —      | 5    |
| 10  | GND    | Signal Ground             | —      | —    |

1. Dual RS-485 mode with echo uses dual RS-422 mode pinout

**Table 20 COM A (RS-485) and COM B (RS-485)**

| Pin | Signal | Function         | In/Out | DB-9 |
|-----|--------|------------------|--------|------|
| 1   | D1-    | COM A - Data (–) | in/out | 1    |
| 2   | D2-    | COM B - Data (–) | in/out | 6    |
| 3   | D1+    | COM A - Data (+) | in/out | 2    |
| 4   | D2+    | COM B - Data (+) | in/out | 7    |
| 5   | —      | Reserved         | —      | 3    |
| 6   | —      | Reserved         | —      | 8    |
| 7   | —      | Reserved         | —      | 4    |
| 8   | —      | Reserved         | —      | 9    |
| 9   | GND    | Signal Ground    | —      | 5    |
| 10  | GND    | Signal Ground    | —      | —    |

## Advanced Digital I/O (aDIO™) Port (CN6)

Connector **CN6** is configured as an aDIO port. aDIO is 12 digital bits configured as 8-bit programmable and 4-bit port programmable I/O, providing any combination of inputs and outputs. Match, event, and strobe interrupt modes mean no more wasting valuable processor time polling digital inputs. Interrupts are generated when the 8-bit programmable digital inputs match a pattern, or on any value change event. Bit masking allows selecting any subgroup of 8 bits. The strobe input latches data into the bit programmable port and generates an interrupt. Refer to *Advanced Digital I/O Ports (aDIO™)*—page 59 for information on programming the aDIO.

**Table 21 aDIO Pinout**

| CN6 Pin | Function | CN6 Pin | Function          |
|---------|----------|---------|-------------------|
| 1       | P0-0     | 2       | P0-1              |
| 3       | P0-2     | 4       | P0-3              |
| 5       | P0-4     | 6       | P0-5              |
| 7       | P0-6     | 8       | P0-7              |
| 9       | strobe 0 | 10      | strobe 1          |
| 11      | P1-0     | 12      | P1-1              |
| 13      | P1-2     | 14      | P1-3              |
| 15      | GND      | 16      | +5 V <sup>1</sup> |

1. Available during standby.

## USB 2.0 Connector (CN17, CN27)

Two USB 2.0 compliant connectors are available on connectors **CN17** and **CN27**. Table 22 provides the pinout of the USB connector.

**Table 22 USB Connector (CN17 and CN27)**

| Pin | Signal     | Function                         | In/Out |
|-----|------------|----------------------------------|--------|
| 1   | VCC1       | Supply +5 V to USB1              | out    |
| 2   | VCC2       | Supply +5 V to USB2              | out    |
| 3   | DATA1–     | Bidirectional data line for USB1 | in/out |
| 4   | DATA2–     | Bidirectional data line for USB2 | in/out |
| 5   | DATA1+     | Bidirectional data line for USB1 | in/out |
| 6   | DATA2+     | Bidirectional data line for USB2 | in/out |
| 7   | GND        | Ground                           | out    |
| 8   | GND        | Ground                           | out    |
| 9   | Shield GND | Shield Ground                    | out    |
| 10  | Shield GND | Shield Ground                    | out    |



**Note** For proper operation at USB 2.0 speeds, be sure to use a cable that is rated for USB 2.0, such as the cable kit supplied by RTD.

Facing the connector pins, the pinout of CN17 is:

|               |     |        |        |      |
|---------------|-----|--------|--------|------|
| 9             | 7   | 5      | 3      | 1    |
| Shield<br>GND | GND | DATA1+ | DATA1– | VCC1 |
| Shield<br>GND | GND | DATA2+ | DATA2– | VCC2 |
| 10            | 8   | 6      | 4      | 2    |

## Ethernet (10/100/1000Base-T and -TX) Connector (CN20, CN30)

This connector provides a 10/100/1000Base-T Ethernet connection. Table 23 provides the pinout of the Ethernet connector. For 1000Base-T, all four pairs are used for transmit and receive.

To use the onboard 10/100/1000 Ethernet controller, Ethernet must be enabled in the BIOS.

When enabled, the multi-color LED will blink to indicate an Ethernet connection. For more information, refer to the *Multi-Color LED* section on page 68.

**Table 23 Ethernet Connector (CN20 and CN30)**

| RJ-45 Pin | 10-Pin DIL Pin | Signal   | Function           |
|-----------|----------------|----------|--------------------|
| 3         | 1              | B+ (RX+) | Receive+ (10/100)  |
| 6         | 2              | B- (RX-) | Receive- (10/100)  |
| 4         | 3              | C+       |                    |
| 5         | 4              | C-       |                    |
| 1         | 5              | A+ (TX+) | Transmit+ (10/100) |
| 2         | 6              | A- (TX-) | Transmit- (10/100) |
| 7         | 7              | D+       |                    |
| 8         | 8              | D-       |                    |
| —         | 9              | AGND     | Ethernet Ground    |
| —         | 10             | AGND     | Ethernet Ground    |

|      |    |    |    |    |
|------|----|----|----|----|
| 9    | 7  | 5  | 3  | 1  |
| AGND | D+ | A+ | C+ | B+ |
| AGND | D- | A- | C- | B- |
| 10   | 8  | 6  | 4  | 2  |

## PCIe/104 Type 2 Bus (CN1 - Top and CN2 - Bottom)

Connectors **CN1** and **CN2** carry the signals of the PCIe/104 PCIe bus. These signals match definitions found in the **PCI/104-Express & PCIe/104 Specification Version 3.1** from the PC/104 Embedded Consortium. Table 24 lists the pinouts of the PC/104-Express bus connector.



**WARNING** Not all PCIe cards are compatible with the PCIe/104 Type 2 connector. Be sure that all of the boards attached to this bus are compatible before powering the system.

**Table 24** PCIe/104 Type 2 Bus Signal Assignments (Top View)<sup>1</sup>

| Pin | Signal                | Signal                | Pin |
|-----|-----------------------|-----------------------|-----|
| 1   | USB_OC#               | PE_RST#               | 2   |
| 3   | +3.3V <sup>2</sup>    | +3.3V <sup>2</sup>    | 4   |
| 5   | USB_1p                | USB_0p                | 6   |
| 7   | USB_1n                | USB_0n                | 8   |
| 9   | GND                   | GND                   | 10  |
| 11  | PEx1_1Tp (CN2 only)   | PEx1_0Tp (CN2 only)   | 12  |
| 13  | PEx1_1Tn (CN2 only)   | PEx1_0Tn (CN2 only)   | 14  |
| 15  | GND                   | GND                   | 16  |
| 17  | PEx1_2Tp (CN2 only)   | PEx1_3Tp (CN2 only)   | 18  |
| 19  | PEx1_2Tn (CN2 only)   | PEx1_3Tn (CN2 only)   | 20  |
| 21  | GND                   | GND                   | 22  |
| 23  | PEx1_1Rp (CN2 only)   | PEx1_0Rp (CN2 only)   | 24  |
| 25  | PEx1_1Rn (CN2 only)   | PEx1_0Rn (CN2 only)   | 26  |
| 27  | GND                   | GND                   | 28  |
| 29  | PEx1_2Rp (CN2 only)   | PEx1_3Rp (CN2 only)   | 30  |
| 31  | PEx1_2Rn (CN2 only)   | PEx1_3Rn (CN2 only)   | 32  |
| 33  | GND                   | GND                   | 34  |
| 35  | PEx1_1Clkp (CN2 only) | PEx1_0Clkp (CN2 only) | 36  |
| 37  | PEx1_1Clkn (CN2 only) | PEx1_0Clkn (CN2 only) | 38  |
| 39  | +5V_STBY              | +5V_STBY              | 40  |
| 41  | PEx1_2Clkp (CN2 only) | PEx1_3Clkp (CN2 only) | 42  |
| 43  | PEx1_2Clkn (CN2 only) | PEx1_3Clkn (CN2 only) | 44  |
| 45  | CPU_DIR               | PWRGOOD               | 46  |
| 47  | Reserved              | Reserved              | 48  |
| 49  | Reserved              | Reserved              | 50  |
| 51  | Reserved              | PSON#                 | 52  |

+5 Volts

**Table 24 PCIe/104 Type 2 Bus Signal Assignments (Top View)<sup>1</sup>**

| Pin | Signal   | Signal              | Pin |
|-----|----------|---------------------|-----|
| 53  | STK0     | STK1                | 54  |
| 55  | GND      | GND                 | 56  |
| 57  | Reserved | Reserved            | 58  |
| 59  | Reserved | Reserved            | 60  |
| 61  | GND      | GND                 | 62  |
| 63  | Reserved | Reserved            | 64  |
| 65  | Reserved | Reserved            | 66  |
| 67  | GND      | GND                 | 68  |
| 69  | Reserved | Reserved            | 70  |
| 71  | Reserved | Reserved            | 72  |
| 73  | GND      | GND                 | 74  |
| 75  | Reserved | Reserved            | 76  |
| 77  | Reserved | Reserved            | 78  |
| 79  | GND      | GND                 | 80  |
| 81  | Reserved | SATA_0Tp (CN1 only) | 82  |
| 83  | Reserved | SATA_0Tn (CN1 only) | 84  |
| 85  | GND      | GND                 | 86  |
| 87  | Reserved | Reserved            | 88  |
| 89  | Reserved | Reserved            | 90  |
| 91  | GND      | GND                 | 92  |
| 93  | Reserved | Reserved            | 94  |
| 95  | Reserved | Reserved            | 96  |
| 97  | GND      | GND                 | 98  |
| 99  | Reserved | Reserved            | 100 |
| 101 | Reserved | Reserved            | 102 |
| 103 | GND      | GND                 | 104 |

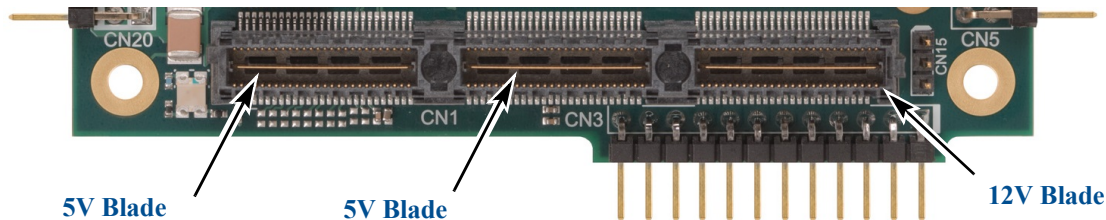
+5 Volts

**Table 24 PCIe/104 Type 2 Bus Signal Assignments (Top View)<sup>1</sup>**

| Pin | Signal   | Signal                       | Pin |
|-----|----------|------------------------------|-----|
| 105 | STK2     | Reserved                     | 106 |
| 107 | GND      | GND                          | 108 |
| 109 | Reserved | Reserved                     | 110 |
| 111 | Reserved | Reserved                     | 112 |
| 113 | GND      | GND                          | 114 |
| 115 | Reserved | Reserved                     | 116 |
| 117 | Reserved | Reserved                     | 118 |
| 119 | GND      | GND                          | 120 |
| 121 | Reserved | Reserved                     | 122 |
| 123 | Reserved | Reserved                     | 124 |
| 125 | GND      | GND                          | 126 |
| 127 | Reserved | Reserved                     | 128 |
| 129 | Reserved | Reserved                     | 130 |
| 131 | GND      | GND                          | 132 |
| 133 | Reserved | SATA_0Rp ( <i>CN1 only</i> ) | 134 |
| 135 | Reserved | SATA_0Rn ( <i>CN1 only</i> ) | 136 |
| 137 | GND      | GND                          | 138 |
| 139 | Reserved | Reserved                     | 140 |
| 141 | Reserved | Reserved                     | 142 |
| 143 | GND      | GND                          | 144 |
| 145 | Reserved | Reserved                     | 146 |
| 147 | Reserved | Reserved                     | 148 |
| 149 | GND      | GND                          | 150 |
| 151 | Reserved | Reserved                     | 152 |
| 153 | Reserved | Reserved                     | 154 |
| 155 | GND      | GND                          | 156 |

+12 Volts<sup>3</sup>

1. Signals marked with (#) are active low.
2. +3.3V is not required for board operation. For power input requirements, see Table 3.
3. +12 Volts not required for operation.



**Figure 6** Locations of 5V and 12V Power Blades

## PCIe/104 Type 2 Compatibility

The PCIe/104 Type 2 connector is compatible with any PCI/104-Express or PCIe/104 peripheral module that does not use the x16 Link. This includes any card that uses the PCIe x1 links, USB, or a power supply. In addition, this connector can be used to add SATA devices to the system.

If a card is installed that is not compatible with the Type 2 connector, the CPU will keep the system in soft-off, and the LED will be Cyan to indicate that there is a Bus Stacking Error.

## PCIe Link Configuration

This cpuModule supports a total of four PCIe x1 links for system expansion, all of which are on the bottom-side PCIe/104 connector, **CN2**. Three of the links on **CN2** are connected directly to the chipset. The fourth link is connected through a PCIe packet switch, which shares the bandwidth of a single x1 link back to the chipset with the onboard PCIe-to-PCI bridge and the gigabit Ethernet controller. Table 25 below shows the configuration of the PCIe x1 links on CN1 and CN2.

**Table 25** CN2 Link Configuration

| Link   | Location          | Connection |
|--------|-------------------|------------|
| Link 3 | Closest to CPU    | Direct     |
| Link 2 |                   | Direct     |
| Link 1 |                   | Direct     |
| Link 0 | Farthest from CPU | Shared     |

## Optional RTC Battery Input (CN13)

The optional RTC battery input is the connection for an external backup battery. This battery is used by the cpuModule when system power is removed in order to preserve the date and time of the real time clock.

Connecting a battery is only required to maintain time when power is completely removed from the cpuModule. A battery is not required for board operation.

**Table 26 Optional RTC Battery Input (CN13)**

| Pin | Signal | Function          |
|-----|--------|-------------------|
| 1   | BAT    | RTC Battery Input |
| 2   | GND    | Ground            |



**WARNING** This optional RTC battery connector (**CN13**) should be left unconnected if the Utility Port 2.0 connector (**CN5**) has a battery connected.

## Reserved (CN15)

Reserved (CN15) should be left unconnected.

## Chapter 4 Using the cpuModule

This chapter provides information for users who wish to develop their own applications programs for the CME34BT cpuModule.

This chapter includes information on the following topics:

*The RTD Enhanced BIOS*—page 54

*Operating System Specific Usage*—page 57

*Non-Standard Serial Port Modes*—page 58

*Advanced Digital I/O Ports (aDIO™)*—page 59

*SATA Controller Configuration*—page 63

*Watchdog Timer Control*—page 64

*Thermal Management*—page 65

*Power Management*—page 66

*Multi-Color LED*—page 68

*Trusted Platform Module*—page 71

*Features and Settings That Can Affect Boot Time*—page 72

*System Recovery*—page 73

# The RTD Enhanced BIOS

The RTD Enhanced BIOS is software that interfaces hardware-specific features of the cpuModule to an operating system (OS). Physically, the BIOS software is stored in a Flash EPROM on the cpuModule. Functions of the BIOS are divided into two parts.

The first part of the BIOS is known as POST (power-on self-test) software, and it is active from the time power is applied until an OS boots (begins execution). POST software performs a series of hardware tests, sets up the machine as defined in Setup, and begins the boot of the OS.

The second part of the BIOS is known as the CORE BIOS. It is the normal interface between cpuModule hardware and the OS which is in control. It is active from the time the OS boots until the cpuModule is turned off. The CORE BIOS provides the system with a series of software interrupts to control various hardware devices.

## Configuring the RTD Enhanced BIOS

The cpuModule Setup program allows you to customize the cpuModule's configuration. Selections made in Setup are stored on the board and are read by the BIOS at power-on.

## Entering the BIOS Setup through the Graphical BIOS Menu

You can run Setup by rebooting the cpuModule and repeatedly pressing the **ESC** key. This will bring to you to a graphical BIOS menu with several options described in the table below.

To enter Setup, navigate to the **Setup Utility** option with the arrow keys, and press **Enter**.



**Table 27 Graphical BIOS Menu Options**

| Key                    | Function                                                                                                                                        |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Continue               | Exit the graphical menu, and proceed with booting                                                                                               |
| Boot Manager           | Select a device to boot to during the current power cycle, and override the selected boot device order in the Setup                             |
| Device Management      | Set the primary video BIOS during the current power cycle, and view a list of the system's detected devices (disk, video, network, input, etc.) |
| Boot From File         | Displays a list of bootable files in the system                                                                                                 |
| Administer Secure Boot | Traditional BIOS Setup                                                                                                                          |

**Table 27 Graphical BIOS Menu Options**

| Key | Function               |
|-----|------------------------|
| SCU | Traditional BIOS Setup |

## Field Selection

To move between fields in Setup, use the keys listed below. When you are finished with Setup, save your changes and exit. The system will automatically reboot.

**Table 28 Setup Keys**

| Key           | Function                               |
|---------------|----------------------------------------|
| <b>F1</b>     | Help                                   |
| →, ←, ↓, ↑    | Move between fields                    |
| <b>F5, F6</b> | Selects next/previous values in fields |
| <b>Enter</b>  | Go to the submenu for the field        |
| <b>ESC</b>    | To previous menu then to exit menu     |
| <b>F8</b>     | Load Setup defaults                    |
| <b>F10</b>    | Save settings and exit Setup           |

## Main Menu Setup Fields

The following is a list of Main Menu Setup fields.

**Table 29 Main Menu Setup Fields**

| Field    | Active Keys                  | Selections                                                                                                                                       |
|----------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Main     | Press <b>Enter</b> to select | Access system information such as the cpuModule's serial number, CPU speed and type, BIOS version, FPGA version, and CMOS time and date settings |
| Advanced | Press <b>Enter</b> to select | Setup advanced RTD cpuModule features, including boot options, aDIO and serial port configuration, and miscellaneous feature control.            |
| Security | Press <b>Enter</b> to select | Setup the supervisor and access password                                                                                                         |
| Boot     | Press <b>Enter</b> to select | Set the system boot sequence                                                                                                                     |
| Power    | Press <b>Enter</b> to select | Set up various power savings modes                                                                                                               |
| Exit     | Press <b>Enter</b> to select | Save or discard changes and exit the BIOS, or load the default BIOS settings                                                                     |



**Note** Future BIOS versions may have slightly different setup menus and options.

**Note** The **Main** page shows the cpuModule's **serial number**, **BIOS Version**, and **FPGA Version**. These numbers can be useful if you need techsupport for your cpuModule.



**WARNING** Incorrect modifications to the BIOS can cause your system to break. If you are unsure of any changes made to the BIOS there is a **Load RTD Defaults** option under the **Exit** tab in the **SCU**. This option restores the BIOS to the default factory settings.

# Operating System Specific Usage

## Windows®

The CME34BT is fully supported under Windows 10 and Windows 7. Both 32 bit and 64 bit editions are supported. Drivers for onboard peripherals such as video and Ethernet are provided on the companion DVD that is shipped with the board. Updated drivers may be available for download from the RTD website. ([www.rtd.com](http://www.rtd.com))

## Linux®

The CME34BT is fully supported with Linux kernel version 3.11 and onward. Earlier versions of the kernel may not contain drivers for all chipset features (most notably SATA), and may not work as expected.

At the time of publication, RTD has validated openSUSE 13.1, openSUSE 13.2, and Ubuntu 15.10 (all 64-bit) on this platform. However, any standard PC Linux distribution with a 3.11 or newer kernel is expected to work.

## UEFI OS Installations

When dealing with UEFI OS installations, the BIOS setting for “EFI/Legacy Device Order” will need to be changed to “EFI devices first”. The default boot order is to boot to “Legacy devices first”, which will render the OS unbootable after an installation.

## Non-Standard Serial Port Modes

It is possible to change the input clock rate for the UARTs of the cpuModule to allow the serial ports to operate at higher speeds than 115,200 bps. This is accomplished by modifying the baud rate multiplier for each serial port.

Non-standard baud rates are supported for all COM port modes of the CME34BT. :

**Table 30 Baud Rate Multiplier Registers**

| Connector | Port  | Address Range (hex) |
|-----------|-------|---------------------|
| CN7 (A)   | COM 1 | EE0–EE1h            |
| CN7 (B)   | COM 3 | EE2–EE3h            |
| CN8 (A)   | COM 2 | EE4–EE5h            |
| CN8 (B)   | COM 4 | EE6–EE7h            |

To set an alternate maximum baud rate for a serial port, write the value which corresponds to the desired multiplier to the port's baud rate multiplier register:

**Table 31 Baud Rate Multipliers**

| Value  | Maximum Baud Rate |            |
|--------|-------------------|------------|
|        | bps               | Multiplier |
| 0x0E28 | 115,200           | 1x         |
| 0x38A0 | 460,800           | 4x         |
| 0x713F | 921,600           | 8x         |
| 0x7AE1 | 1,000,000         | 8.7x       |



**Note** When using the non-standard high speed serial port modes, it is highly recommended to use hardware flow control, whenever possible.

## Advanced Digital I/O Ports (aDIO™)

This board supports 12 bits of TTL/CMOS compatible digital I/O (TTL signaling). These I/O lines are grouped into two ports, Port 0 and Port 1. Port 0 is bit programmable; Port 1 is byte programmable. Port 0 supports RTD's Advanced Digital Interrupt modes. The three modes are strobe, match and event. Strobe mode generates an interrupt and latches Port 0 when the strobe input transitions from low to high. Match mode generates an interrupt when an 8-bit pattern is received in parallel that matches the match mask register. Event mode generates an interrupt when a change occurs on any bit. In any mode, masking can be used to monitor selected lines.

When the CPU boots, all digital I/O lines are programmed as inputs, meaning that the digital I/O line's initial state is undetermined. If the digital I/O lines must power up to a known state, an external 10 k-Ohm resistor must be added to pull the line high or low.

The 8-bit control read/write registers for the digital I/O lines are located from I/O address EC0h to EC3h. These registers are written to zero upon power up. From EC0h to EC3h, the name of these registers are **Port 0 data**, **Port 1 data**, **Multi-Function**, and **DIO-Control** register.



***Note** While the 16-pin aDIO connector is pin-for-pin compatible with previous generations of RTD cpuModules (such as the Montevina series), the hardware is slightly different, and requires that writes and reads to/from the aDIO ports happen no more than once per millisecond. Migrating legacy software to the E3800 Series that utilizes RTD's Advanced Digital I/O may require that you add this delay prior to all reads and writes.*

***Note** RTD provides drivers that support the aDIO interface on popular operating systems. RTD recommends using these drivers instead of accessing the registers directly.*

### Digital I/O Register Set

**Table 32 Port 0 Data I/O Address EC0h**

| D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   |
|------|------|------|------|------|------|------|------|
| P0.7 | P0.6 | P0.5 | P0.4 | P0.3 | P0.2 | P0.1 | P0.0 |

Port 0 Data register is a read/write bit direction programmable register. A particular bit can be set to input or output. A read of an input bit returns the value of port 0. A read of an output bit returns the last value written to Port 0. A write to an output bit sends that value to port 0.

**Table 33 Port 1 Data I/O Address EC1h**

| D7       | D6       | D5       | D4       | D3   | D2   | D1   | D0   |
|----------|----------|----------|----------|------|------|------|------|
| Reserved | Reserved | Reserved | Reserved | P1.3 | P1.2 | P1.1 | P1.0 |

Port 1 Data register is a read/write byte direction programmable register. A read on this register when it is programmed to input will read the value at the aDIO connector. A write on this register when it is programmed as output will write the value to the aDIO connector. A read on this register when it is set to output will read the last value sent to the aDIO connector.

**Table 34 Multi-Function I/O Address EC2h**

| D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|----|----|----|----|----|----|----|----|
|----|----|----|----|----|----|----|----|

The multi-function register is a read/write register whose contents are set by the DIO-Control register. See the DIO-Control register description for a description of this register.

**Table 35 DIO-Control I/O Address EC3h—Read Access**

| D7                               | D6                                                                      | D5                               | D4                                                                                          | D3 | D2                            | D1                                                                                                                        | D0 |
|----------------------------------|-------------------------------------------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------|----|-------------------------------|---------------------------------------------------------------------------------------------------------------------------|----|
| Strobe 0 Status<br>0 = no strobe | Digital IRQ Status<br>0 = no digital interrupt<br>1 = digital interrupt | Strobe 1 Status<br>0 = no strobe | Digital IRQ Mode<br>00 = Disabled<br>01 = strobe Mode<br>10 = event mode<br>11 = match mode |    | Port 1 Direction<br>0 = input | Multi-Function Register Select<br>00 = clear mode<br>01 = port 0 direction<br>10 = mask register<br>11 = compare register |    |
|                                  |                                                                         |                                  |                                                                                             |    |                               |                                                                                                                           |    |

**Table 36 DIO-Control I/O Address EC3h—Write Access**

| D7       | D6 | D5 | D4                                                                                          | D3 | D2                            | D1                                                                                                                        | D0 |
|----------|----|----|---------------------------------------------------------------------------------------------|----|-------------------------------|---------------------------------------------------------------------------------------------------------------------------|----|
| Reserved |    |    | Digital IRQ Mode<br>00 = Disabled<br>01 = strobe Mode<br>10 = event mode<br>11 = match mode |    | Port 1 Direction<br>0 = input | Multi-Function Register Select<br>00 = clear mode<br>01 = port 0 direction<br>10 = mask register<br>11 = compare register |    |
|          |    |    |                                                                                             |    |                               |                                                                                                                           |    |

| Table 37 Multi-Function at Address EC2h <sup>1</sup> |                     |     |     |     |     |     |     |     |
|------------------------------------------------------|---------------------|-----|-----|-----|-----|-----|-----|-----|
| read/write                                           | 00 clear            | X   | X   | X   | X   | X   | X   | X   |
| 0 in, 1 out                                          | 01 Port 0 direction | I/O | I/O | I/O | I/O | I/O | I/O | I/O |
| 0 no mask, 1 mask                                    | 10 DIO mask         | M7  | M6  | M5  | M4  | M3  | M2  | M1  |
| read/write                                           | 11 compare          | C7  | C6  | C5  | C4  | C3  | C2  | C1  |
|                                                      |                     | C0  |     |     |     |     |     |     |

1. Contents based on bits D0 and D1 of DIO-Control.

### Clear Register:

A read to this register Clears the IRQs and a write to this register sets the DIO-Compare, DIO- Mask, DIO-Control, Port 1, and Port 0 to zeros. A write to this register is used to clear the board.

### Port 0 Direction Register:

Writing a zero to a bit in this register makes the corresponding pin of the aDIO connector an input. Writing a one to a bit in this register makes the corresponding pin of the aDIO connector an output.

### Mask Register:

Writing a zero to a bit in this register will not mask off the corresponding bit in the DIO-Compare register. Writing a one to a bit in this register masks off the corresponding bit in the DIO-Compare register. When all bits are masked off the aDIOs comparator is disabled. This condition means Event and Match mode will not generate an interrupt. This register is used by Event and Match modes.

### Compare Register:

A Read/Write register used for Match Mode. Bit values in this register that are not masked off are compared against the value on Port 0. A Match or Event causes bit 6 of DIO-Control to be set and if the aDIO is in Advanced interrupt mode, the Match or Event causes an interrupt.

Port 1 Data register is a read/write byte direction

## Interrupts

In order to use an interrupt with aDIO, the interrupt must first be selected in the BIOS setup utility under **Advanced, RTD Advanced Options, RTD aDIO, aDIO Interrupt**. The Digital I/O can use interrupts 3, 5, 6, 10, and 11. To configure the aDIO interrupt, navigate to the BIOS Setup option, the “Advanced”, “RTD aDIO”, and “aDIO Interrupt” menu options. The BIOS will automatically reserve the selected interrupt so that it is not assigned to PCI devices. Then, select the appropriate interrupt mode in the DIO Control register.

### Advanced Digital Interrupts

There are three Advanced Digital Interrupt modes available. These three modes are Event, Match, and Strobe. The use of these three modes is to monitor state changes at the aDIO connector. Interrupts are enabled by writing to the **Digital IRQ Mode** field in the **DIO-Control** register.

#### Event Mode

When this mode is enabled, Port 0 is latched into the DIO-Compare register. The aDIO circuitry includes deglitching logic. To enter Event mode, set bits [4:3] of the DIO-Control register to “10”.

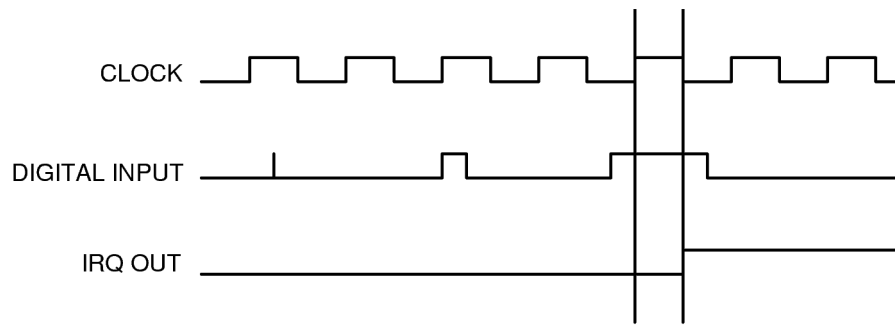
#### Match Mode

When this mode is enabled, Port 0 is latched into the DIO-Compare register. The aDIO circuitry includes deglitching logic. To enter Match mode, set bits [4:3] of the DIO-Control register to “11”.



**Note** Make sure bits [4:3] are set *BEFORE* writing the DIO-Compare register. If you do not set them first, the contents of the DIO-Compare register could be lost because the Event mode latches in Port 0 into the DIO-Compare register.

---



**Figure 7 aDIO Match Mode**

### **Strobe Mode**

Another interrupt mode supported by aDIO is Strobe mode. This allows the strobe pin of the DIO connector to trigger an interrupt. A low to high transition on the strobe pin will cause an interrupt request. The request will remain high until the Clear Register is read from. Additionally, the Compare Register latched in the value at Port 0 when the Strobe pin made a low to high transition. No further strobes will be available until a read of the Compare Register is made. You must read the Compare Register, and then clear interrupts so that the latched value in the compare register is not lost. To enter Strobe mode, set bits [4:3] of the DIO-Control register to “01”.

# SATA Controller Configuration

The CME34BT has two SATA links. One SATA link resides on the top PCIe/104 Type 2 connector (**CN1**), and the second SATA link connects to the onboard SATA storage. Both may be enabled or disabled through the BIOS setup.

## SATA Port Mappings, by Connector

Table 38 shows the SATA port configuration of the CME34BT cpuModule:

**Table 38 SATA Port Layout**

| Designator | Physical Position, Index | E3800 Port Index | Channel Assignment (Legacy IDE Mode) |
|------------|--------------------------|------------------|--------------------------------------|
| <b>CN1</b> | SATA PCIe link 0         | 0                | Primary Master                       |
| <b>n/a</b> | Onboard SATA storage     | 1                | Primary Slave                        |

## Onboard SATA Storage

The cpuModule was designed to be used in embedded computing applications. In such environments, rotating media like hard disks and floppy disks are not very desirable. It is possible to eliminate rotating storage devices by placing your operating system and application software into the cpuModule's onboard SATA storage.

The onboard SATA storage supports write protection. Write protection is disabled by default and can be enabled with a menu option in the BIOS setup.

## SATA Links on the PCIe/104 Type Connectors

Both SATA links are connected to the Intel E3800 CPU's SATA controller which can be configured for either ACHI or IDE mode through the BIOS . However, the operating system must support the selected mode for the device to operate correctly. The default configuration for the controller is ACHI mode.

When in IDE mode, the user has an additional option to change the IDE emulation mode to Native Mode or Legacy Mode. When the SATA controller is configured for IDE mode, Native Mode is the default IDE emulation mode.

### Native IDE Mode

Native Mode allows more flexibility than Legacy Mode, as it permits the system to modify the resources used by the SATA controller. When in Native Mode, the SATA controller only requires a single IRQ. Unlike Legacy Mode, this IRQ may be changed by the user or the operating system for better distribution of the system IRQs. When IRQs in the system are more evenly distributed, interrupt latency is minimized. The base address of the controller may also be modified.

### Legacy IDE Mode

When in this mode, the controller will be fixed to use two interrupts: IRQs 14 and 15. Similarly, the I/O address of the controller will be fixed in the system. When in Legacy Mode, only a primary and secondary channel may be used in the system.

# Watchdog Timer Control

The cpuModule includes a Watchdog Timer, which provides protection against programs “hanging”, or getting stuck in an execution loop where they cannot respond correctly. The watchdog timer consists of a counter, a reset generator, and an interrupt generator. When the counter reaches the interrupt time-out, it can generate an interrupt. When the counter reaches the reset time-out, the system is reset. The counter is “refreshed,” or set back to zero by reading from a specific register. The watchdog can also be put into an “inactive” state, in which no resets or interrupts are generated.

The ability to generate an interrupt allows the application to gracefully recover from a bad state. For example, consider a system that has a reset time-out of 2 seconds, interrupt time-out of 1 second, and the watchdog timer is refreshed every 0.5 seconds. If something goes wrong, an interrupt is generated. The Interrupt service routine then attempts to restart the application software. If it is successful, the application is restarted in much less time than a full reboot would require. If it is not successful, the system is rebooted.

Due to system latency, it is recommended that the Watchdog be refreshed at about half of the reset time-out period, or half of the interrupt time-out period, whichever is applicable.

## Register Description

The Advanced Watchdog Timer has a Setup Register and a Runtime Register. The Setup Register is set by the BIOS, and can be adjusted by entering the BIOS Setup Utility, and going to “Advanced”, “RTD Advanced Options”, “RTD Miscellaneous Feature Control”. The Setup Register may also be read by the driver to determine if the Watchdog is enabled, and the interrupt and base address that it is using.



**Note** Enabling the watchdog timer in the BIOS does not actually arm it. The watchdog timer can be armed by accessing I/O address E85h, as explained below.

**Table 39 Advanced Watchdog Setup Register E8Bh**

| D7       | D6 | D5 | D4 | D3                       | D2 | D1                                                                             | D0 |
|----------|----|----|----|--------------------------|----|--------------------------------------------------------------------------------|----|
| Reserved |    |    |    | <b>WDT_IRQ</b>           |    | <b>Reg_Enable</b>                                                              |    |
|          |    |    |    | Select Interrupt for WDT |    | 0 = Watchdog timer is disabled and Runtime Register will not appear in I/O map |    |
|          |    |    |    | 000 = Disabled           |    | 1 = Watchdog Timer is enabled.                                                 |    |
|          |    |    |    | 001 = IRQ5               |    | Runtime Register will appear in I/O map                                        |    |
|          |    |    |    | 010 = Reserved           |    |                                                                                |    |
|          |    |    |    | 011 = IRQ10              |    |                                                                                |    |
|          |    |    |    | 100 = IRQ11              |    |                                                                                |    |
|          |    |    |    | 101 = Reserved           |    |                                                                                |    |
|          |    |    |    | 110 = IRQ3               |    |                                                                                |    |
|          |    |    |    | 111 = IRQ6               |    |                                                                                |    |

**Table 40 Advanced Watchdog Runtime Register E85h**

| D7                                                            | D6                                  | D5       | D4 | D3                            | D2 | D1                        | D0 |
|---------------------------------------------------------------|-------------------------------------|----------|----|-------------------------------|----|---------------------------|----|
| <b>WDT_Active</b>                                             | <b>WDT_IRQ_En</b>                   | Reserved |    | <b>WDT_IRQ_Ti</b>             |    | <b>WDT_RST_Ti</b>         |    |
| 0 = Watchdog timer is disabled.                               | 0 = Watchdog interrupt is disabled. |          |    | Select Interrupt time for WDT |    | Select Reset time for WDT |    |
| 1 = Watchdog is armed and can generate resets and interrupts. | 1 = Watchdog interrupt is enabled.  |          |    | 00 = 0.25s                    |    | 00 = 2.00s                |    |
|                                                               |                                     |          |    | 01 = 0.50s                    |    | 01 = 0.50s                |    |
|                                                               |                                     |          |    | 10 = 0.75s                    |    | 10 = 0.75s                |    |
|                                                               |                                     |          |    | 11 = 1.00s                    |    | 11 = 1.00s                |    |

Reading the Runtime Register also refreshes the watchdog timer.

# Thermal Management

The cpuModule has several thermal features which can be used to monitor and control the board's temperature when extreme operating conditions are prevalent.

## Thermal Warning

The cpuModule includes a Thermal Warning mechanism which will assert a signal when the processor's temperature has reached or exceeded 100 C. At this time, a bit will be set in the reset and event status register to indicate that this temperature has been entered. Although the LED has turned white and a thermal warning bit has been set, the CPU will maintain the current power and operating state.

For more information on the reset status register, refer to *Reset and Event Status Register*—page 69.

## ACPI-Enforced Thermal Protection

When the maximum thermal junction temperature of the processor is reached, the system is at risk of malfunctioning, and the cpuModule may become permanently damaged.

For operating system with ACPI support, the system offers two mechanisms to help prevent the processor from reaching its maximum thermal junction temperature -- the **Critical Trip Point** and the **Passive Trip Point**. Both of these temperatures may be configured in the BIOS Setup.

### Critical Trip Point

The Critical Trip Point controls the temperature at which an ACPI operating system will shut down the system. Upon entering the Critical Trip state, the operating system will issue a command to the processor, and the CME34BT will immediately transition to the **S5** (Soft-Off) ACPI power state.

To exit the Critical Trip state (and S5), the CPU temperature must be reduced to a point that the internal thermal sensors are below the Critical Trip Point. Next, a power cycle or RSM reset (deepest reset) is required. The reset button may be configured as an RSM reset (default BIOS setting) in the BIOS setup utility.

For more information on the S5 power state, refer to *Advanced Configuration and Power Interface (ACPI)*—page 66.

### Passive Trip Point

The Passive Trip Point controls the temperature at which an ACPI operating system will begin to throttle the processor.

## Further Temperature Reduction

The cpuModule's temperature is directly related to power consumption. Reducing the power consumption of the cpuModule will have an effect on the cpuModule's temperature. Suggested methods for reducing the cpuModule's power consumption can be found in the *Power Management* section on page 66.

# Power Management

The CME34BT cpuModule supports various powering mechanisms which allow the cpuModule to monitor power consumption and temperature, and achieve minimal power consumption states. These unique features include thermal monitoring and thermal throttling, as well as ACPI low power modes. Various wake options are also available to resume normal system power.

## Advanced Configuration and Power Interface (ACPI)

The cpuModule supports several different ACPI low power modes, including the S3, S4, and S5 sleeping states. These suspend modes are described below:

- **S3 (Suspend to RAM):** When the system wakes from this mode, operating systems allow applications to resume where they left off, as the state of the application is preserved in memory.
- **S4 (Hibernate):** When the system enters this state, the operating system will save the current state of applications and relevant data to disk, thus allowing the system RAM to be powered down.
- **S5 (Soft-Off):** The system is in a soft off state, and must be rebooted when it wakes.

## Power Button Modes

The soft power button input of the Utility Port 2.0 connector (**CN5**) can be configured by the operating system as a suspend button (transition to S3) or as soft power button (transition to S5). Consult your operating system documentation for information on how to configure it. The power button will always cause a transition to S5 if pressed for 4 seconds or longer, without interaction from the operating system.

The soft power button of the Utility Port 2.0 connector (**CN5**) is the only mechanism to wake the system from S4 and S5.



**Note** When the reset button of the Utility Port 2.0 connector is configured as a power button (via the BIOS setup utility), it inherits the ACPI suspend and resume features of the power button.

## Low-Power Wake Options

The cpuModule supports several methods of waking from a low power state.

- **Resume from USB:** Operating systems that support S3 permit waking the system from the USB interfaces. Common USB wake methods include insertion of a USB device, a USB keyboard stroke, or movement from another USB device, such as mouse.
- **Resume on RTC Alarm / Timeout:** The RTC Alarm allows the system to turn on at a certain time every day.

**Table 41 Supported ACPI Resume/Wake Mechanisms**

| Resume/Wake Mechanism             | S3 | S4 | S5 |
|-----------------------------------|----|----|----|
| Wake from USB                     | Y  | —  | —  |
| Wake from RTC Alarm / Timeout     | Y  | —  | —  |
| Power Button Input ( <b>CN5</b> ) | Y  | Y  | Y  |

## AT vs. ATX Power Supplies

Both AT and ATX power supplies may be used with the CME34BT cpuModule, however AT power supplies do not provide any standby power to the cpuModule. When an AT power supply is used to power the system, low power modes that require a standby power to wake the system will not be fully supported.

ATX power supplies do provide a standby power, thus allowing the system to utilize all low power modes supported by the hardware. When an ATX supply is used to power the cpuModule, lower power modes can be achieved. During these low power modes, the standby power from the ATX power supply provides power to a small circuit on the CPU, which is used to watch for a system wake event.

## ATX Power Supply Signals

The PCIe/104 Type 2 Bus connectors (**CN1** & **CN2**) provide two ATX style signals., +5V Standby and PSON#. The +5V Standby rail is used to power certain parts of the cpuModule when the main power supply is turned off, i.e. during Suspend-to-RAM (S3), Hibernate (S4), or Soft-Off (S5) power modes. The PSON# signal is an active low open-drain output that signals the power supply to turn on. Use of these signals allows the power consumption to drop to below 1W during standby modes, and still enable any of the wake events.

## Reducing Power Consumption

In addition to the CPU's low power modes, power consumption can further be reduced by making some modifications to the BIOS setup. When the following features are disabled in the BIOS, the CPU's power consumption will decrease:

- **Ethernet**
- **USB Ports**
- **SATA Controller**
- **Serial Ports**
- **Multi-Color LED**

## Multi-Color LED

The CME34BT has a Multi-Color LED which can be enabled or disabled in the BIOS setup screen. The color of the LED indicates the status of the board, as shown in Table 42.

**Table 42 LED Colors**

| Color                | Description                                         |
|----------------------|-----------------------------------------------------|
| Green                | Normal Operation                                    |
| Blue                 | SATA Activity                                       |
| Red <sup>1</sup>     | cpuModule is in reset                               |
| Yellow (Red + Green) | cpuModule is in Standby                             |
| White (R+G+B)        | cpuModule is approaching thermal limit <sup>2</sup> |
| Cyan (Blue + Green)  | Ethernet Link at 100 Mbps or Bus Stacking Error     |
| Magenta (Blue + Red) | Ethernet Link at 1000 Mbps                          |
| Blink                | Ethernet Activity                                   |

1. If power is applied to the cpuModule while jumper **JP5** is installed, the LED will be red. This does not indicate that the board is in reset
2. The LED will remain White until the system is shut down.

The LED can also be controlled manually by writing to I/O Port EA7h, as shown in Table 43 and Table 44.

**Table 43 Multi-Color LED I/O Address EA5h**

| D7       | D6       | D5       | D4       | D3       | D2 | D1              | D0 |
|----------|----------|----------|----------|----------|----|-----------------|----|
| Reserved | Reserved | Reserved | Reserved | Reserved |    | Multi-Color LED |    |

The following table lists the color displayed and the value written.

**Table 44 Manual LED Colors**

| I/O Port EA7h Value | Color                                      |
|---------------------|--------------------------------------------|
| 0x00                | Automatic (see Table 42)                   |
| 0x08                | Off (will reduce system power consumption) |
| 0x09                | Blue                                       |
| 0x0A                | Green                                      |
| 0x0B                | Cyan (Green + Blue)                        |
| 0x0C                | Red                                        |
| 0x0D                | Magenta (Red + Blue)                       |
| 0x0E                | Yellow (Red + Green)                       |
| 0x0F                | White (Red + Green + Blue)                 |

## Reset and Event Status Register

The cpuModule has several different signals on board which can cause a system reset. If a reset occurs, the reset status register can be used to see which reset (or resets) have been asserted on the cpuModule. These resets may be cleared by the user.

Similar to resets, there are several signals on the cpuModule which indicate that a special event has occurred. These may also be monitored and cleared by the user.

- **Examine Resets and Events:** Reading from I/O ports 0xED0h through 0xED3 will indicate if a reset has been asserted or if an event has occurred. If a 1 is read, the corresponding reset has been asserted. If a 0 is read from the bit, the reset has not been asserted. For events
- **Clear Reset and Event:** Each reset and event can be cleared by writing a 1 to the selected bit of I/O port 0xED0h.

**Table 45 Reset and Event Status I/O Address ED0h - Read Access**

| D7                                                    | D6                                               | D5                                                               | D4                                             | D3                                                     | D2                                               | D1                                                               | D0                                                         |
|-------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------|------------------------------------------------|--------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------|
| <b>Sleep S4</b><br>1 = reset asserted<br>0 = no reset | <b>PwrGood 1.8V Always</b><br>1 = reset asserted | <b>PwrGood 1.0V Always</b><br>1 = reset asserted<br>0 = no reset | <b>PwrGood 5V Always</b><br>1 = reset asserted | <b>RSM Reset</b><br>1 = reset asserted<br>0 = no reset | <b>PwrGood 1.5V Always</b><br>1 = reset asserted | <b>PwrGood 3.3V Always</b><br>1 = reset asserted<br>0 = no reset | <b>Main 5V Power</b><br>1 = reset asserted<br>0 = no reset |

**Table 46 Reset and Event Status I/O Address ED1h - Read Access**

| D7                                                         | D6                                                          | D5                                                    | D4                                               | D3                                                        | D2                                                        | D1                                                               | D0                                           |
|------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------|----------------------------------------------|
| <b>PwrGood 1.05V</b><br>1 = reset asserted<br>0 = no reset | <b>Pwr Good Vcore</b><br>1 = reset asserted<br>0 = no reset | <b>Sleep S3</b><br>1 = reset asserted<br>0 = no reset | <b>PwrGood 1.2V Always</b><br>1 = reset asserted | <b>PwrGood 1.0V</b><br>1 = reset asserted<br>0 = no reset | <b>PwrGood Vgfx</b><br>1 = reset asserted<br>0 = no reset | <b>PwrGood 1.9V Always</b><br>1 = reset asserted<br>0 = no reset | <b>Memory Pwr Good</b><br>1 = reset asserted |

**Table 47 Reset and Event Status I/O Address ED2h - Read Access**

| D7                                                    | D6                                           | D5                                                          | D4              | D3                                                              | D2                                                        | D1                                                        | D0                                                         |
|-------------------------------------------------------|----------------------------------------------|-------------------------------------------------------------|-----------------|-----------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------|
| <b>WDT Trip</b><br>1 = reset asserted<br>0 = no reset | <b>Thermal Warning</b><br>1 = reset asserted | <b>Platform Reset</b><br>1 = reset asserted<br>0 = no reset | <b>Reserved</b> | <b>Mem. Term. Voltage</b><br>1 = reset asserted<br>0 = no reset | <b>PwrGood 3.3V</b><br>1 = reset asserted<br>0 = no reset | <b>PwrGood 1.8V</b><br>1 = reset asserted<br>0 = no reset | <b>PwrGood 1.35V</b><br>1 = reset asserted<br>0 = no reset |

**Table 48 Reset and Event Status I/O Address ED3h - Read Access**

| D7              | D6              | D5                                                        | D4                                                           | D3              | D2              | D1                                                          | D0                                                         |
|-----------------|-----------------|-----------------------------------------------------------|--------------------------------------------------------------|-----------------|-----------------|-------------------------------------------------------------|------------------------------------------------------------|
| <b>Reserved</b> | <b>Reserved</b> | <b>ATX Pwr Good</b><br>1 = reset asserted<br>0 = no reset | <b>Reserved</b>                                              | <b>Reserved</b> | <b>Reserved</b> | <b>Soft Power Off</b><br>1 = reset asserted<br>0 = no reset | <b>Utility Reset</b><br>1 = reset asserted<br>0 = no reset |
|                 |                 |                                                           | <b>Bus Stack Error</b><br>1 = reset asserted<br>0 = no reset |                 |                 |                                                             |                                                            |

# Trusted Platform Module

The CME34BT cpuModule includes a Trusted Platform Module (TPM) that is fully compliant to the Trusted Computing Group (TCG) TPM version 2.0 specification.

TPM stands for “Trusted Platform Module”. The TPM is a controller on your cpuModule that provides tamper resistant storage using hardware-based cryptography. The TPM can be used to store passwords, certificates, and keys to encrypted file systems. The TPM protects the data on your drives from being read should your system or drive be stolen. An un-encrypted drive can be removed from your system and connected to another, bypassing the need for a user account and password. With a TPM encrypted drive the only way to access the data is by logging into the system as a user with access to the files containing your sensitive information. TPM 2.0 controllers can generate and store [AES-256-GCM-Sha256](#) encryption keys to ensure your data can not be decrypted. As such, it is highly recommended to save a copy of any file system encryption keys either on paper or a USB drive in the event your device experiences a hardware failure. Without this key your data will be lost with NO chance of recovery. For best results, this key should never be saved locally, even temporarily, as the key may leave a lasting trace on flash or magnetic drives.

TPM chips are still vulnerable to software attacks, malware, and viruses. Once the system has booted and a user is logged in, the encrypted drives are unlocked with the keys from the TPM, allowing any software attacks full access to any important data and even the keys within the TPM. As such, it is still highly recommended to maintain firewalls, anti-virus, and any other internet security methods.

Before using the TPM chip it must be enabled and cleared in the Security Tab of the BIOS. Windows must also be installed in UEFI mode to allow TPM access. UEFI boot mode can be set in the Boot tab of the bios. For Debian-based Linux distributions refer to “TPM Implementation White Paper”. For Red Hat Linux refer to guides found on the Red Hat Customer Portal for your version.

For more information, refer to the Trusted Platform Module version 2.0 specification.

**Table 49 Additional References:**

| Title                          | Description                                                               | Source                                                                                                                                                                        |
|--------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TPM Specifications             | TPM specs, versions, updates                                              | <a href="https://trustedcomputinggroup.org/resource/tpm-library-specification/">https://trustedcomputinggroup.org/resource/tpm-library-specification/</a>                     |
| TPM Summary and Operation      | An in-depth white paper on how the TPM works.                             | <a href="https://trustedcomputinggroup.org/resource/trusted-platform-module-tpm-summary/">https://trustedcomputinggroup.org/resource/trusted-platform-module-tpm-summary/</a> |
| TPM Implementation White Paper | An in-depth white paper on how to set up TPM drivers in Windows and Linux | Contact RTD Techsupport                                                                                                                                                       |
| Red Hat Customer Portal        | Customer portal for Red Hat Subscribers                                   | <a href="https://access.redhat.com/">https://access.redhat.com/</a>                                                                                                           |

## Features and Settings That Can Affect Boot Time

The boot time of a system is dependent upon numerous system settings as well as devices attached to a system. This section addresses some devices and settings that can increase or decrease a system's boot time.

### Boot Device Order

The BIOS contains a list of devices to try booting from. If you wish to boot to a particular device (for example, a hard drive), make sure that it is first in the boot order. This will speed up boot times.

### Add-On Cards With BIOS Extensions

Some add-on cards have an integrated BIOS extension. The most common examples are SCSI controllers and network cards with boot ROMs. During POST, the BIOS executes the card's extension code. This extension code is third-party code, which is beyond RTD's control. The BIOS extension will most likely increase the boot time. Exactly how much it increases boot time will depend on the particular card and firmware version.

### VGA Controller

VGA controllers have a VGA BIOS that must be initialized during POST. It can take some time to initialize the VGA BIOS. Exactly how long will depend on the particular VGA controller and BIOS version.

### Hard Drive Type

During Hard Drive initialization, each device must be probed. Some devices take longer to probe. 2.5-inch hard drives tend to take longer than 3.5-inch ones, because they spin at a lower RPM.

### Monitor Type

Some monitors take a while to power on. Desktop flat panels are especially slow. This does not affect the actual boot time of the CPU. However, the CPU may boot before the monitor powers on.

# System Recovery

## Reset Button Recovery

The CME34BT provides several methods for recovering from an incorrectly configured system. In order to enter the recovery mode, follow the steps below:

1. Remove power from the system, including standby power.
2. Press and hold the reset button attached to the Utility Port 2.0 connector.
3. Apply power to the system while continuing to hold the reset button.
4. Wait the amount of time shown in Table 50 for the desired recovery mode.
5. Release the reset button, allowing the system to boot.

**Table 50 Reset Button Recovery Modes**

| Hold Time <sup>1</sup> | Mode                                                                 |
|------------------------|----------------------------------------------------------------------|
| 0-4 seconds            | No recovery mode. System will stay in reset while button is pressed. |
| 4-8 seconds            | Load Default BIOS Settings                                           |
| 8-12 seconds           | Serial POST Code Output                                              |

1. To assist with timing while the reset button is pushed, the LED will blink OFF at 5 second intervals.

## Load Default BIOS Settings

Loading BIOS defaults allows recovery from an incorrectly configured display device, incorrect boot options, and many other incorrect settings. It is also a good starting point when making BIOS changes. After restoring defaults, the BIOS settings should be reviewed and modified as needed.

The default BIOS can be restored either by using Reset Button Recovery, or the “Load RTD Defaults” option in the BIOS.

## Serial Power-On-Self-Test (POST) Code Output

The POST Codes represent a series of events that take place in a system during the Power On Self Test. If the POST fails, the system will not boot as expected. Knowing which POST code the failure occurred may help system debug.

This recovery mode configures serial port connector **CN7** as single RS-232, and sends the POST codes on the transmit pin. The port settings are 115kbps, 8 bits, no parity, one stop bit. When using this recovery mode, the POST codes can be logged on another computer running terminal software. Contact RTD technical support for more details.



**Note** POST Codes over the serial port (**CN7**) may also be enabled by installing **JP5** and **JP6** prior to power cycling the cpuModule. For more information, see Jumper Settings and Locations on page 76.



# Appendix A Hardware Reference

This appendix provides information on CME34BT cpuModule hardware, including:

*Jumper Settings and Locations*—page 76

*Onboard PCI/PCIe Devices*—page 77

*Physical Dimensions*—page 78

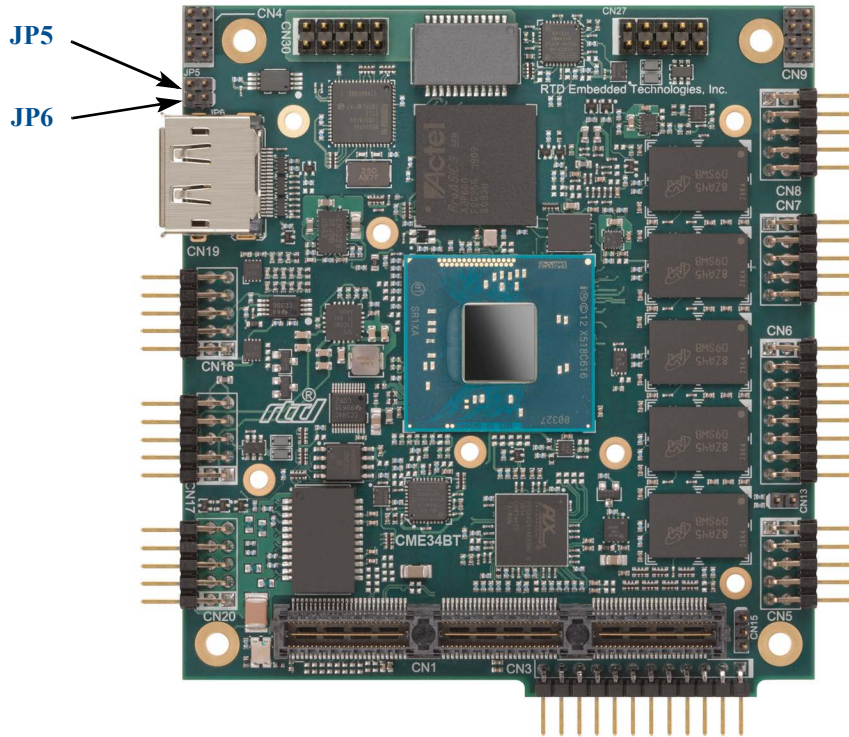
*Heatsink Dimensions*—page 79

*Flat-Heatspreader Dimensions*—page 81

## Jumper Settings and Locations

Many cpuModule options are configured by positioning jumpers. Jumpers are labeled on the board as **JP** followed by a number.

Figure 8 shows the jumper locations that are used to configure the cpuModule. Table 51 lists the jumpers and their settings.



**Figure 8 CME34BT Jumper Locations (top side)**

**Table 51 CME34BT Jumpers**

| Jumper           | Pins | Function                                                                                                                                                                                                                                                                                               | Default |
|------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| <b>JP5</b>       | 2    | Reserved                                                                                                                                                                                                                                                                                               | open    |
| <b>JP6</b>       | 2    | Reserved                                                                                                                                                                                                                                                                                               | open    |
| <b>JP5 + JP6</b> | 2    | Used to configure serial port connector <b>CN7</b> as single RS-232, and sends the POST codes on the transmit pin. See <i>Serial Power-On-Self-Test (POST) Code Output</i> on page 73<br><b>pins 1–2 (JP5 and JP6):</b> Enable POST Codes over Serial Port <b>CN7</b><br><b>open:</b> Normal operation | open    |

## Onboard PCI/PCIe Devices

The CME34BT cpuModule has several onboard PCI/PCIe devices, all of which are listed in the table below. This table shows a typical configuration, and the actual devices may change based on BIOS settings.

**Table 52 Onboard PCI/PCIe Devices**

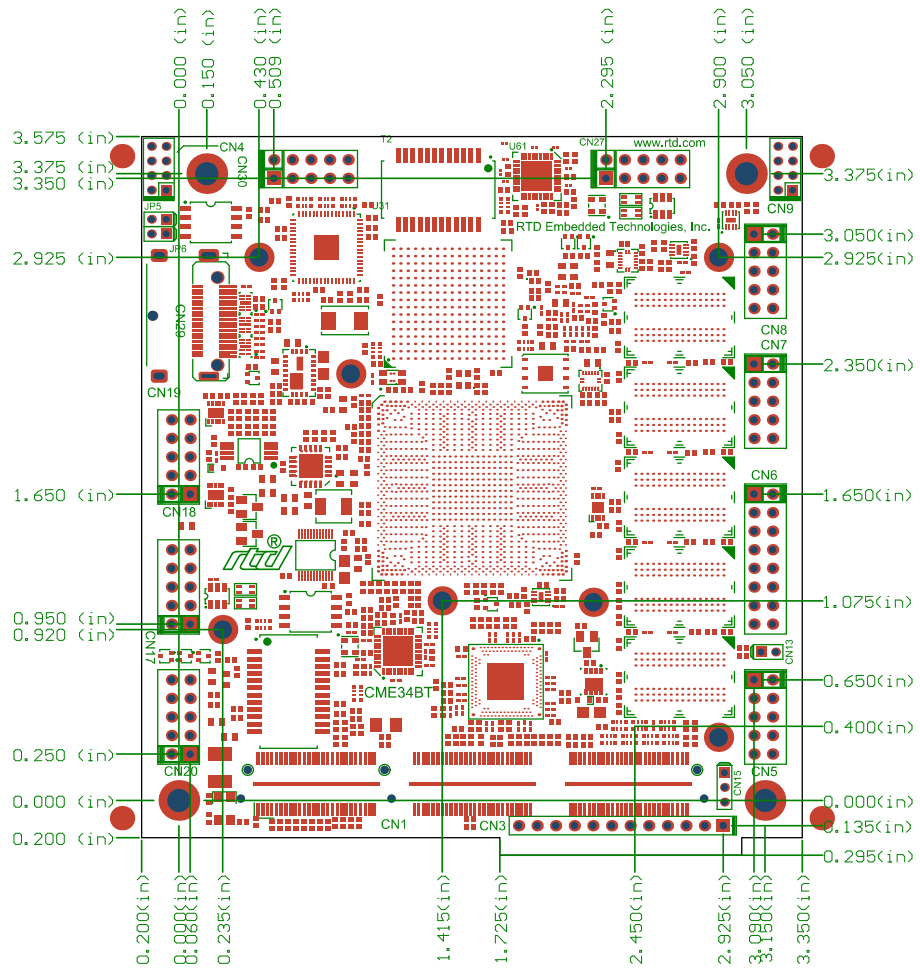
| Device ID | Vendor ID | Description                                                 |
|-----------|-----------|-------------------------------------------------------------|
| 0F00      | 8086      | Host bridge                                                 |
| 0F04      | 8086      | HD Audio Controller                                         |
| 0F31      | 8086      | Video Controller                                            |
| 0F23      | 8086      | SATA Controller (AHCI Mode) <sup>1</sup>                    |
| 0F18      | 8086      | Intel TXE (Trusted Execution Engine)                        |
| 0F48      | 8086      | PCI Express Root Port 1                                     |
| 0F4A      | 8086      | PCI Express Root Port 2                                     |
| 0F4C      | 8086      | PCI Express Root Port 3                                     |
| 0F4E      | 8086      | PCI Express Root Port 4                                     |
| 0F34      | 8086      | USB Controller (EHCI)                                       |
| 0F1C      | 8086      | Platform Controller Unit - LPC Bridge                       |
| 0F12      | 8086      | Platform Controller Unit - SMBus Port                       |
| 8605      | 10B5      | PCI Express Packet Switch <sup>2</sup>                      |
| 8112      | 10B5      | PCI Express to PCI Bridge                                   |
| 1531      | 8086      | Ethernet Controller - Intel WGI210IT Series (CN20 and CN30) |

1.Can change if the BIOS is configured for IDE mode instead of AHCI.

2.This device may appear up to five times in PCI configuration space, depending on how many PCIe links are active.

## Physical Dimensions

Figure 9 shows the mechanical dimensions of the CME34BT cpuModule.



**Figure 9 CME34BT Physical Dimensions ( $\pm 0.005$  inches)**

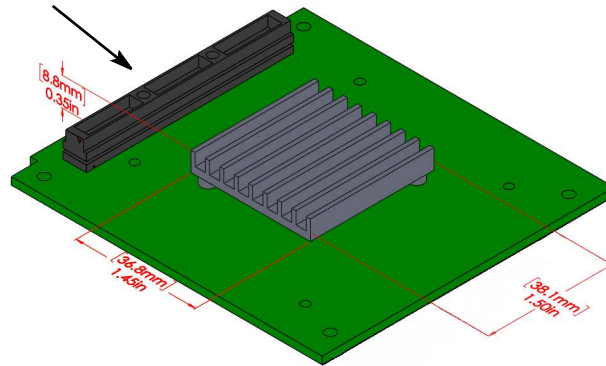
## Heatsink Dimensions

To maximize the thermal performance of the cpuModule while keeping the total board weight to a minimum, various heatsinks are offered with each ordering part number. Use the table below to correlate the heatsink figure to each ordering part number.

**Table 53 CME34BT cpuModule Model Options**

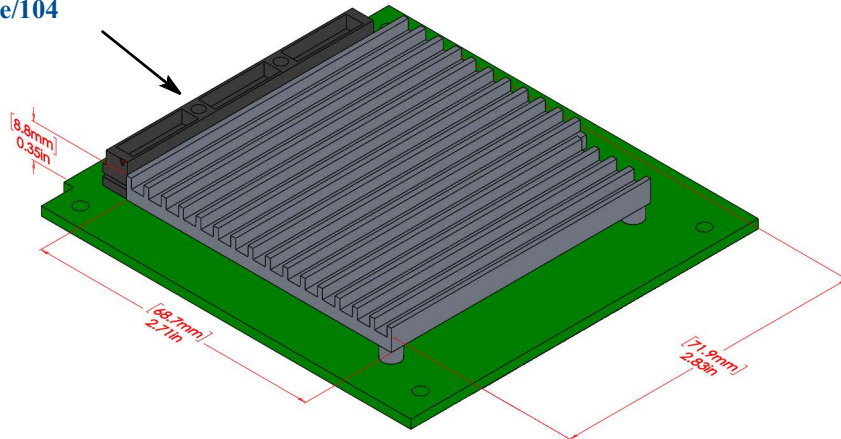
| Part Number               | Core Frequency | Stack Height     | Heatsink  |
|---------------------------|----------------|------------------|-----------|
| CME34BTS1460HR-4096/S32GX | 1.46 GHz       | 0.600" (15.24mm) | Figure 10 |
| CME34BTD1330HR-4096/S32GX | 1.33 GHz       | 0.600" (15.24mm) | Figure 11 |
| CME34BTQ1910HR-4096/S32GX | 1.91 GHz       | 0.866" (22mm)    | Figure 12 |

Edge near  
PCIe/104



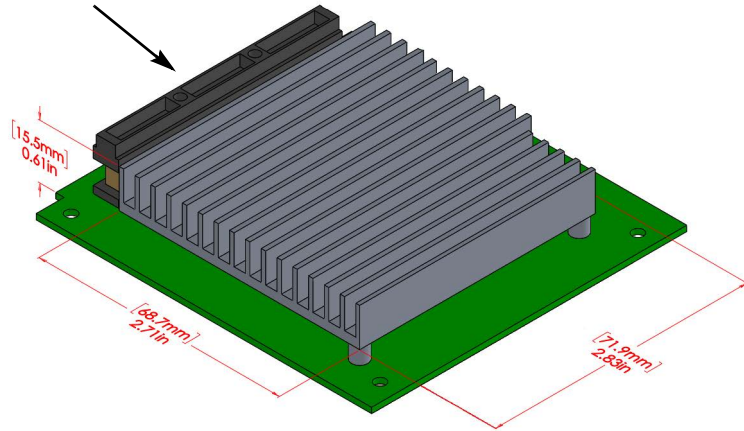
**Figure 10 CME34BTS1460 Heatsink**

Edge near  
PCIe/104



**Figure 11 CME34BTD1330 Heatsink**

Edge near  
PCIe/104



**Figure 12 CME34BTQ1910 Heatsink**

## Flat-Heatspreader Dimensions

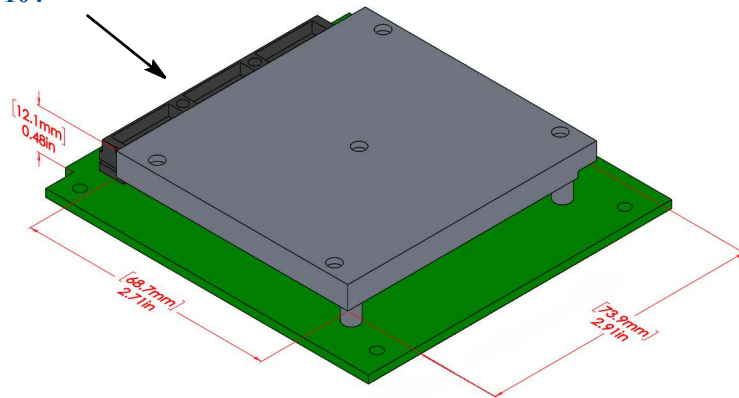
For cooling solutions where it is desirable to mount the cpuModule to a flat surface, the CME34BT may be ordered with a flat heatspreader instead of the standard passive heatsink. Each flat-heatspreader has five mounting locations which use 6-32 UNC x 0.207" long threaded insert.

Use the table below to correlate the heatsink figure to each ordering part number.

**Table 54 CME34BT cpuModule Model Options**

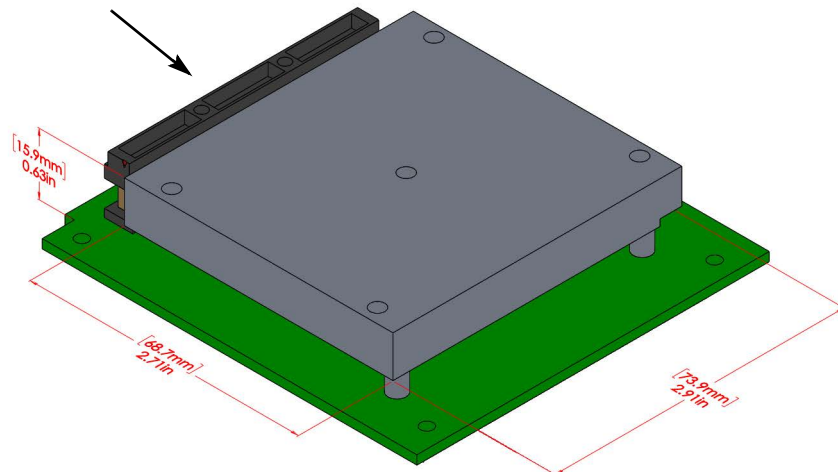
| Part Number                | Core Frequency | Stack Height     | Heatspreader |
|----------------------------|----------------|------------------|--------------|
| CME34BTS1460HR-4096/S32GXF | 1.46 GHz       | 0.600" (15.24mm) | Figure 13    |
| CME34BTD1330HR-4096/S32GXF | 1.33 GHz       | 0.600" (15.24mm) | Figure 13    |
| CME34BTQ1910HR-4096/S32GXF | 1.91 GHz       | 0.866" (22mm)    | Figure 14    |

Edge near  
PCIe/104



**Figure 13 0.600" (15.24mm) Flat-Heatspreader**

Edge near  
PCIe/104



**Figure 14 0.866" (22mm) Flat-Heatspreader**



## Appendix B Troubleshooting

Many problems you may encounter with operation of your CME34BT cpuModule are due to common errors. This appendix includes the following sections to help you get your system operating properly.

*Common Problems and Solutions*—page 84

*Troubleshooting a PC/104 System*—page 85

*How to Obtain Technical Support*—page 86

## Common Problems and Solutions

Table 55 lists some of the common problems you may encounter while using your CME34BT cpuModule, and suggests possible solutions.

If you are having problems with your cpuModule, review this table before contacting RTD Technical Support.

**Table 55 Troubleshooting**

| Problem                                   | Additional Symptoms                                       | Possible Cause(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cpuModule does not boot (no video output) | LED is red (in reset)                                     | Input power is inadequate. Refer to Chapter 1, <i>Electrical Characteristics</i> . Verify input voltages are correct. If power is supplied via CN3, check for cabling issues.<br><br>Malfunctioning peripheral card. Remove all peripheral cards from the system.                                                                                                                                                                                                                                                                                                           |
|                                           | LED is yellow (in standby)                                | cpuModule is in “Soft Off” due to operating system shutdown. Press power button to wake.<br><br>ATX power supply is not turning on. Check jumpers on power supply. Verify PS_ON# signal.                                                                                                                                                                                                                                                                                                                                                                                    |
|                                           | LED is cyan (bus stacking error)                          | See Chapter 3,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                           | LED is green (normal operation)                           | cpuModule may be booted, but video is not connected properly. Attach a keyboard and verify the operation of the Num Lock and Caps Lock LEDs. If they respond as expected, the cpuModule is booted. Check the video cable connections.<br><br>Video output may have been disabled in the BIOS. Try loading BIOS defaults per Chapter 4, <i>System Recovery</i> .<br><br>If a PCI device is installed, the slot selection switch/jumpers may not be set correctly.<br><br>Peripheral card may be interfering with boot sequence. Remove all peripheral cards from the system. |
| cpuModule reboots unexpectedly            | cpuModule is running Windows                              | By default, Windows will automatically reboot when a Bug check (Blue Screen) is triggered. Disable this feature in the Control Panel. This will allow you to see the error and identify the root cause.                                                                                                                                                                                                                                                                                                                                                                     |
|                                           | cpuModule is powered via Auxiliary Power Connector (CN3)  | Power cabling issues. Verify all pins on CN3 are reliably connected with proper gauge wires.<br><br>Power supply noise or current limiting. Check power pins with an oscilloscope and verify they remain in range.<br><br>Power supply may not respond quickly enough to changes in power consumption. Switch to a different power supply (a PCI Express power supply is recommended)                                                                                                                                                                                       |
|                                           | Peripheral card(s) installed                              | Peripheral card(s) may be overloading the power supply. Reduce the system to just the cpuModule and power supply and see if the problem re-appears. (Note that CN3 is intended for powering the cpuModule only.)                                                                                                                                                                                                                                                                                                                                                            |
| cpuModule does not boot to USB device     | USB device has a high power draw (e.g. DVD or hard drive) | Some USB devices are not fully compliant and draw too much current. Consider switching to a self-powered USB device with its own AC/DC power supply.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Date/time not saved when system is off    | No RTC battery                                            | A battery must be attached to the Utility Port 2.0 connector (CN5) to maintain date/time when main power is removed. For a list of alternate battery inputs, refer to <i>Watchdog Timer Control</i> —page 64.                                                                                                                                                                                                                                                                                                                                                               |

**Table 55 Troubleshooting (cont'd)**

| Problem                                | Additional Symptoms | Possible Cause(s)                                                                                                    |
|----------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------|
| System performance lower than expected | LED is white        | cpuModule is overheating and the processor is throttling. Increase the cooling (more airflow, larger heatsink, etc). |

## Troubleshooting a PC/104 System

If you have reviewed the preceding table and still cannot isolate the problem with your CME34BT cpuModule, please try the following troubleshooting steps. Even if the resulting information does not help you find the problem, it will be very helpful if you need to contact technical support.

1. **Simplify the system.** Remove items one at a time and see if one particular item seems to cause the problem.
2. **Swap components.** Try replacing items in the system one-at-a-time with similar items.

## How to Obtain Technical Support

If after following the above steps, you still cannot resolve a problem with your CME34BT cpuModule, please gather the following information:

- cpuModule model, BIOS version, and serial number
- List of all boards in system
- List of settings from cpuModule Setup program
- Printout of autoexec.bat and config.sys files (if applicable)
- Description of problem
- Circumstances under which problem occurs

Then contact RTD Technical Support:

Phone: 814-234-8087

Fax: 814-234-5218

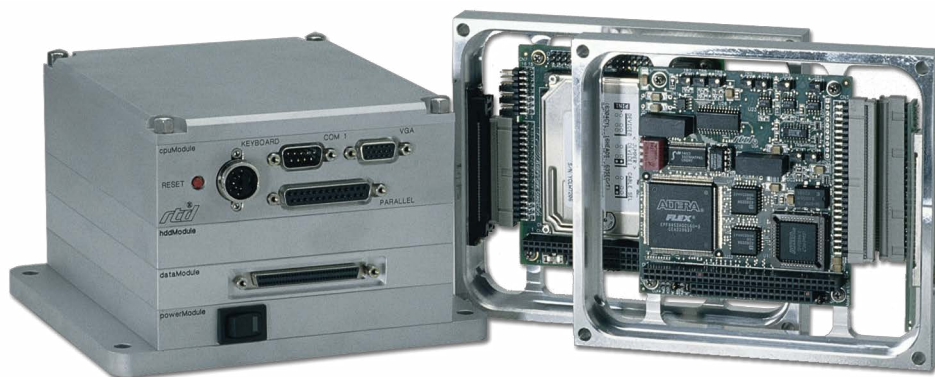
E-mail: [techsupport@rtd.com](mailto:techsupport@rtd.com)

## Appendix C IDAN™ Dimensions and Pinout

Like all other RTD PC/PCI-104 and PCIe/104 modules, cpuModules can be packaged in Intelligent Data Acquisition Node (IDAN) frames, which are milled aluminum frames with integrated heat sinks and heat pipes for fanless operation. RTD modules installed in IDAN frames are called building blocks. IDAN building blocks maintain the simple but rugged stacking concept of PC/PCI-104 and PCIe/104. Each RTD module is mounted in its own IDAN frame and all I/O connections are brought to the walls of each frame using standard PC connectors.

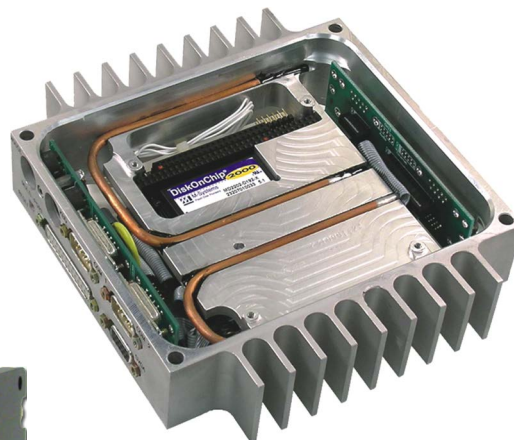
On the CME34BT, no connections are made from module to module internal to the system other than through the PCIe/104 Type 2 bus, enabling quick interchangeability and system expansion without hours of rewiring and board redesign.

The CME34BT cpuModule can also be purchased as part of a custom-built RTD HiDAN™ or HiDAN*plus*™ High Reliability Intelligent Data Acquisition Node. This appendix provides the dimensions and pinouts of the CME34BT installed in an IDAN frame. Contact RTD for more information on high reliability IDAN, HiDAN, and HiDAN*plus* PC/PCI-104 systems.



**IDAN**—Adhering to the PC/PCI-104 and PCIe/104 stacking concept, IDAN allows you to build a customized system with any combination of RTD modules.

**IDAN Heat Pipes**—Advanced heat pipe technology maximizes heat transfer to heat



**HiDANplus**—Integrating the modularity of IDAN with the ruggedization of HiDAN, HiDANplus enables connectors on all system frames, with signals running between frames

## IDAN Contents

The IDAN-CME34BT contains both a CME34BT cpuModule and a **SATA34106** storage module which provides a 2.5" SATA interface to the first SATA link on the CPU's top PCIe/104 Type 2 connector (CN1).

Also inside the IDAN-CME34BT unit is a CMOS battery which is connected to the battery input connection of the cpuModule's Utility Port 2.0 connector. The frame of the IDAN-CME34BT brings out the cpuModule's multi-color LED as well as the cpuModule's reset button. While a power button is not provided on the exterior of the frame, the reset button is configurable as a power button through the BIOS Setup utility.

For additional flexibility, the 25-pin "D" connector which provides connections to the cpuModule's aDIO connector (CN6) includes connections to several pins on the Utility Port 2.0 connector, which include the reset button input, power button input, and RTC battery input.

For more information on cpuModule's battery input and reset button input on the Utility Port 2.0 connector, refer to the *Utility Port 2.0 Connector (CN5)* on page 32.

## IDAN Dimensions

The IDAN frame for the CME34BT is shown in the figure below:

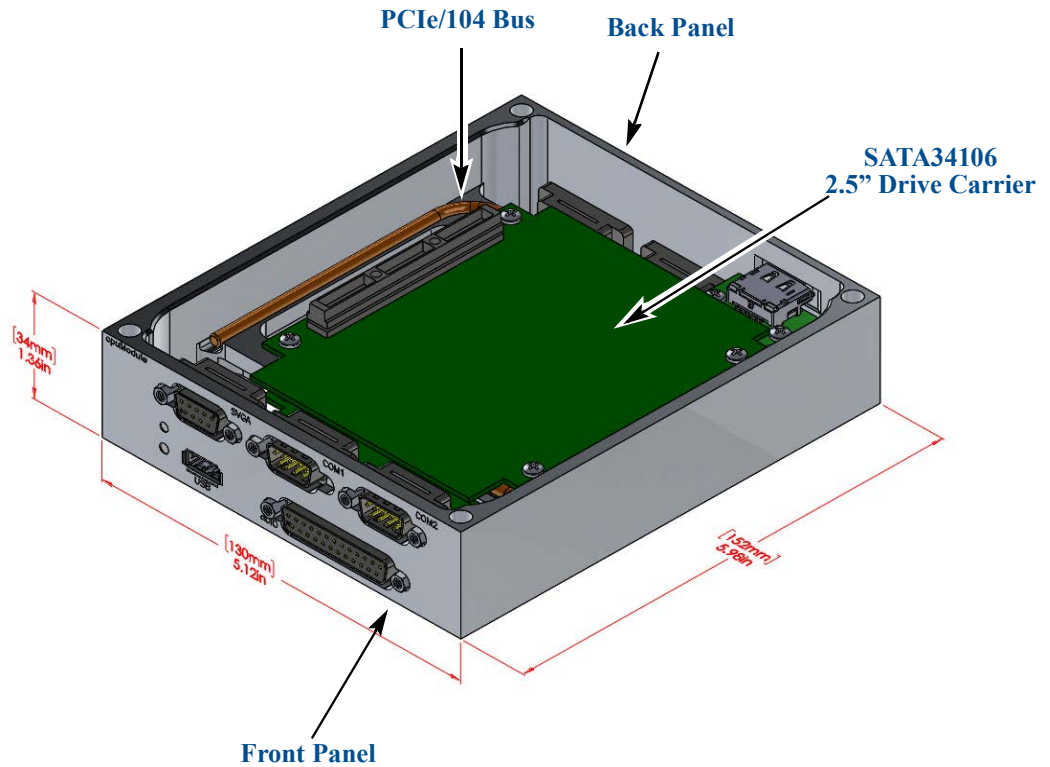
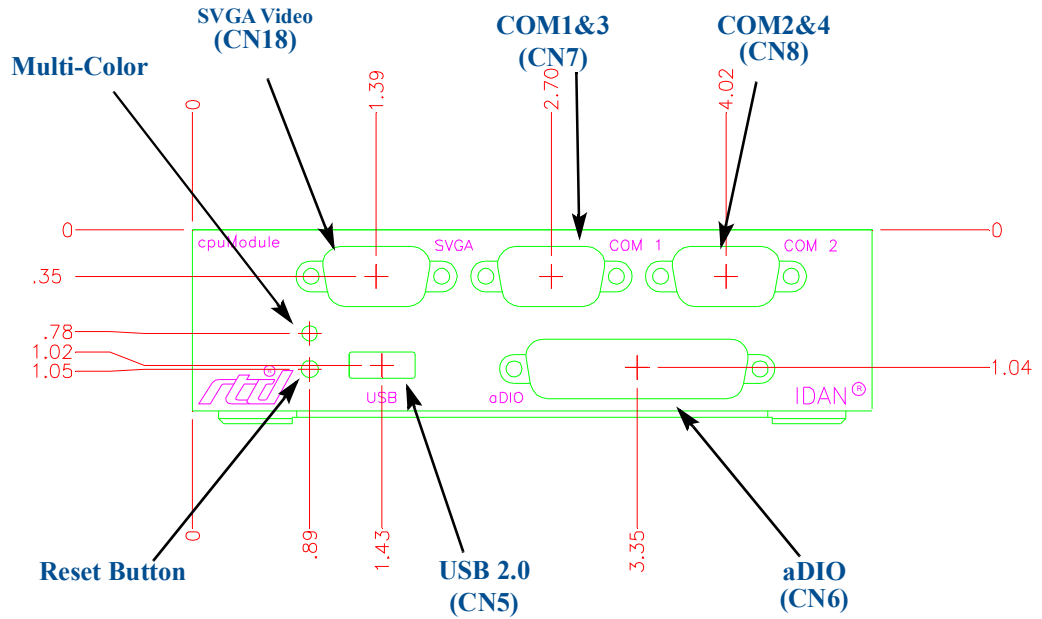


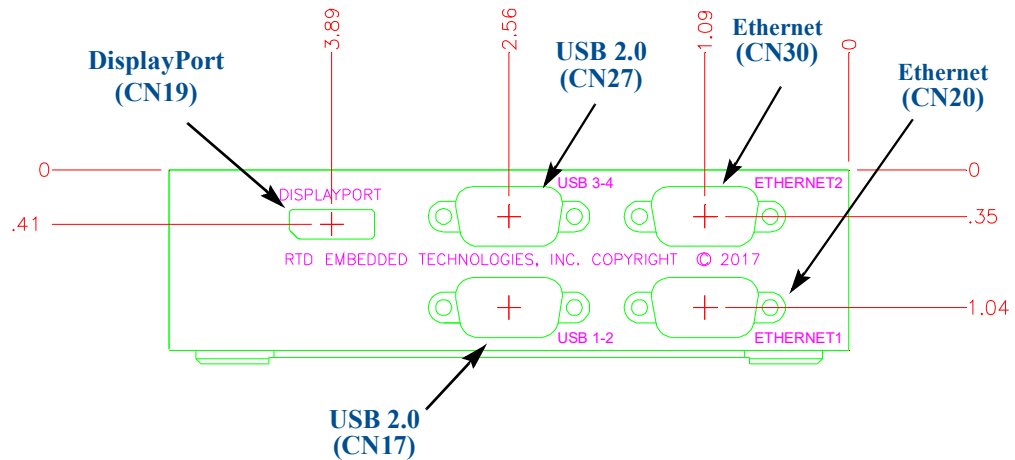
Figure 15 IDAN-CME34BT  
Figure 16

## IDAN Connector Locations

The diagrams below show the connector locations for the headers of the CME34BT as they are brought out on the front and back panels of the IDAN-CME34BT. For a full description of each connector on the CME34BT, refer to *Connecting the cpuModule* on page 27.



**Figure 17 IDAN-CME34BT Front Panel**



**Figure 18 IDAN-CME34BT Back Panel**

**Table 56 IDAN-CME34BT Connectors**

| <b>Designator</b>    | <b>Function</b>             | <b>Connector Description</b>   | <b>IDAN Panel Connector</b> | <b>Mating Connector</b> |
|----------------------|-----------------------------|--------------------------------|-----------------------------|-------------------------|
| <b>CN5</b>           | USB Port (Utility Port 2.0) | USB Type A                     | Series “A” Receptacle       | Series “A” Plug         |
| <b>CN6</b>           | aDIO                        | 25-pin D (female)              | Adam Tech DB25SD            | Adam Tech DB25PD        |
| <b>CN7</b>           | Serial Port 1 (COM1&3)      | 9-pin D (male)                 | Adam Tech DE09PD            | Adam Tech DE09SD        |
| <b>CN8</b>           | Serial Port 2 (COM2&4)      | 9-pin D (male)                 | Adam Tech DE09PD            | Adam Tech DE09SD        |
| <b>CN17 and CN27</b> | USB 2.0                     | 9-pin D (male)                 | Adam Tech DE09PD            | Adam Tech DE09SD        |
| <b>CN18</b>          | Video (SVGA)                | 15-pin high-density D (female) | Adam Tech HDT15SD           | Adam Tech HDT15PD       |
| <b>CN19</b>          | DisplayPort                 | DisplayPort (female)           | Molex 47272 series          | Molex 68783 series      |
| <b>CN20 and CN30</b> | Ethernet                    | 9-pin D (male)                 | Adam Tech DE09PD            | Adam Tech DE09SD        |

## External I/O Connections

**Table 57 COM1/COM2 (Single Port Mode)— 9-Pin D Connector (male)**

| CPU Pin | IDAN Pin (DB-9) | Single RS-232 | Single RS-422 | Single RS-485 |
|---------|-----------------|---------------|---------------|---------------|
| 1       | 1               | DCD           | TXD-          | D-            |
| 2       | 6               | DSR           | —             | —             |
| 3       | 2               | RXD           | TXD+          | D+            |
| 4       | 7               | RTS           | —             | —             |
| 5       | 3               | TXD           | RXD+          | —             |
| 6       | 8               | CTS           | —             | —             |
| 7       | 4               | DTR           | RXD-          | —             |
| 8       | 9               | RI            | —             | —             |
| 9       | 5               | GND           | GND           | GND           |

**Table 58 COM1/COM2 (Dual Port Mode)— 9-Pin D Connector (male)**

| CPU Pin | IDAN Pin (DB-9) | Dual RS-232 | Dual RS-422 | Dual RS-485 |
|---------|-----------------|-------------|-------------|-------------|
| 1       | 1               | DCD1        | TXD1-       | D1-         |
| 3       | 2               | RXD1        | TXD1+       | D1+         |
| 5       | 3               | TXD1        | RXD1+       | —           |
| 7       | 4               | RTS1        | RXD1-       | —           |
| 9       | 5               | GND         | GND         | GND         |
| 2       | 6               | CTS1        | TXD2-       | D2-         |
| 4       | 7               | TXD2        | TXD2+       | D2+         |
| 6       | 8               | RXD2        | RXD2+       | —           |
| 8       | 9               | RI1         | RXD2-       | —           |

**Table 59 aDIO — 25-Pin D Connector (female)**

| IDAN Pin # | aDIO Port | CPU Pin # (CN6) | CPU Pin # (CN5) |
|------------|-----------|-----------------|-----------------|
| 1          | P0-0      | 1               | —               |
| 2          | P0-2      | 3               | —               |
| 3          | P0-4      | 5               | —               |
| 4          | P0-6      | 7               | —               |
| 5          | Strobe 0  | 9               | —               |
| 6          | P1-0      | 11              | —               |
| 7          | P1-2      | 13              | —               |

**Table 59 aDIO — 25-Pin D Connector (female)**

| IDAN Pin # | aDIO Port                   | CPU Pin #<br>(CN6) | CPU Pin #<br>(CN5) |
|------------|-----------------------------|--------------------|--------------------|
| 8          | GND                         | 15                 | —                  |
| 9          | reserved                    | —                  | —                  |
| 10         | reserved                    | —                  | —                  |
| 11         | reserved                    | —                  | —                  |
| 12         | Soft Power Button Input     | —                  | 5                  |
| 13         | GND (for RTC Battery Input) | —                  | 7                  |
| 14         | P0-1                        | 2                  | —                  |
| 15         | P0-3                        | 4                  | —                  |
| 16         | P0-5                        | 6                  | —                  |
| 17         | P0-7                        | 8                  | —                  |
| 18         | Strobe 1                    | 10                 | —                  |
| 19         | P1-1                        | 12                 | —                  |
| 20         | P1-3                        | 14                 | —                  |
| 21         | +5 V                        | 16                 | —                  |
| 22         | reserved                    | —                  | —                  |
| 23         | reserved                    | —                  | —                  |
| 24         | Push-Button Reset Input     | —                  | 3                  |
| 25         | RTC Battery Input           | —                  | 9                  |

**Table 60 SVGA — 15-Pin High Density D Connector (female)**

| IDAN Pin # | Signal   | Function            | CPU Pin # |
|------------|----------|---------------------|-----------|
| 1          | Red      | Red Analog Output   | 4         |
| 2          | Green    | Green Analog Output | 6         |
| 3          | Blue     | Blue Analog Output  | 8         |
| 4          | Reserved | Reserved            | —         |
| 5          | GND      | Ground              | 9         |
| 6          | GND      | Ground              | 9         |
| 7          | GND      | Ground              | 9         |
| 8          | GND      | Ground              | 10        |
| 9          | +5 V     | + 5 Volts           | 7         |
| 10         | GND      | Ground              | 10        |
| 11         | Reserved | Reserved            | —         |
| 12         | DDC Data | Monitor data        | 5         |
| 13         | HSYNC    | Horizontal Sync     | 2         |
| 14         | VSYNC    | Vertical Sync       | 1         |
| 15         | DDC CLK  | Monitor Clock       | 3         |

**Table 61 DisplayPort Connector**

| Pin | Signal | Standard Function            | DVI / HDMI mode <sup>1</sup>              | In/Out |
|-----|--------|------------------------------|-------------------------------------------|--------|
| 1   | LN0+   | Main Link, Lane 0 (positive) | TMDS Channel 2 (positive)                 | out    |
| 2   | GND    | Ground                       | Ground                                    | out    |
| 3   | LN0-   | Main Link, Lane 0 (negative) | TMDS Channel 2 (negative)                 | out    |
| 4   | LN1+   | Main Link, Lane 1 (positive) | TMDS Channel 1 (positive)                 | out    |
| 5   | GND    | Ground                       | Ground                                    | out    |
| 6   | LN1-   | Main Link, Lane 1 (negative) | TMDS Channel 1 (negative)                 | out    |
| 7   | LN2+   | Main Link, Lane 2 (positive) | TMDS Channel 0 (positive)                 | out    |
| 8   | GND    | Ground                       | Ground                                    | out    |
| 9   | LN2-   | Main Link, Lane 2 (negative) | TMDS Channel 0 (negative)                 | out    |
| 10  | LN3+   | Main Link, Lane 3 (positive) | TMDS Clock (positive)                     | out    |
| 11  | GND    | Ground                       | Ground                                    | out    |
| 12  | LN3-   | Main Link, Lane 3 (negative) | TMDS Clock (negative)                     | out    |
| 13  | CFG1   | Configuration Pin 1          | Cable Adapter Detect                      | out    |
| 14  | CFG2   | Configuration Pin 2          | Consumer Electronics Control <sup>2</sup> | out    |
| 15  | AUX+   | Auxiliary Channel (positive) | DDC Clock                                 | in/out |
| 16  | GND    | Ground                       | Ground                                    | out    |
| 17  | AUX-   | Auxiliary Channel (negative) | DDC Data                                  | in/out |
| 18  | HPD    | Hot Plug Detect              | Hot Plug Detect                           | in     |
| 19  | DPG    | Return for DPV               | Return for DPV                            | out    |
| 20  | DPV    | +3.3V DC Power               | +3.3V DC Power                            | out    |

1.Requires special passive adapter

2.HDMI mode only

Facing DisplayPort connector, the pinout is:

|     |      |      |      |      |      |      |      |      |      |
|-----|------|------|------|------|------|------|------|------|------|
| 19  | 17   | 15   | 13   | 11   | 9    | 7    | 5    | 3    | 1    |
| DPG | AUX- | AUX+ | CFG1 | GND  | LN2- | LN2+ | GND  | LN0- | LN0+ |
| DPV | HPD  | GND  | CFG2 | LN3- | LN3+ | GND  | LN1- | LN1+ | GND  |
| 20  | 18   | 16   | 14   | 12   | 10   | 8    | 6    | 4    | 2    |

**Table 62 USB — 9-Pin D Connector (male)**

| IDAN Pin # | Signal     | Function      | Mode         | CPU Pin # |
|------------|------------|---------------|--------------|-----------|
| 1          | VCC1       | +5 V to USB1  | output       | 1         |
| 2          | Data USB1– | USB1 Data–    | input/output | 3         |
| 3          | Data USB1+ | USB1 Data+    | input/output | 5         |
| 4          | GND        | Ground        | —            | 7         |
| 5          | Shield GND | Shield Ground | —            | 9         |
| 6          | VCC2       | +5 V to USB2  | output       | 2         |
| 7          | Data USB2– | USB2 Data–    | input/output | 4         |
| 8          | Data USB2+ | USB2 Data+    | input/output | 6         |
| 9          | GND        | Ground        | —            | 8         |

**Table 63 Ethernet — 9-Pin D Connector (male)**

| IDAN Pin # | RJ-45 Pin | Signal   | CPU Pin # |
|------------|-----------|----------|-----------|
| 1          | 3         | B+ (RX+) | 1         |
| 2          | 4         | C+       | 3         |
| 3          | 1         | A+ (TX+) | 5         |
| 4          | 7         | D+       | 7         |
| 5          | —         | Ground   | 9         |
| 6          | 6         | B- (RX–) | 2         |
| 7          | 5         | C-       | 4         |
| 8          | 2         | A- (TX–) | 6         |
| 9          | 8         | D-       | 8         |





# **Appendix D Additional Information**

## **Application Notes**

RTD offers many application notes that provide assistance with the unique feature set of the CME34BT cpuModule. For the latest application notes, refer to the RTD website.

## **Drivers and Example Programs**

To obtain the latest versions of drivers and example programs for this cpuModule, refer to the RTD website.

## **Interrupt Programming**

For more information about interrupts and writing interrupt service routines, refer to the following book:

Interrupt-Driven PC System Design  
by Joseph McGivern  
ISBN: 0929392507

## **Serial Port Programming**

For more information about programming serial port UARTs, consult the following book:

Serial Communications Developer's Guide  
by Mark Nielson  
ISBN: 0764545701

## **PC/104 Specifications**

A copy of the latest PC/104 specifications can be found on the webpage for the PC/104 Embedded Consortium:

<http://www.pc104.org>



## Appendix E Limited Warranty

RTD Embedded Technologies, Inc. warrants the hardware and software products it manufactures and produces to be free from defects in materials and workmanship for one year following the date of shipment from RTD Embedded Technologies, Inc. This warranty is limited to the original purchaser of product and is not transferable.

During the one year warranty period, RTD Embedded Technologies will repair or replace, at its option, any defective products or parts at no additional charge, provided that the product is returned, shipping prepaid, to RTD Embedded Technologies. All replaced parts and products become the property of RTD Embedded Technologies. Before returning any product for repair, customers are required to contact the factory for a Return Material Authorization number.

This limited warranty does not extend to any products which have been damaged as a result of accident, misuse, abuse (such as: use of incorrect input voltages, improper or insufficient ventilation, failure to follow the operating instructions that are provided by RTD Embedded Technologies, “acts of god” or other contingencies beyond the control of RTD Embedded Technologies), or as a result of service or modification by anyone other than RTD Embedded Technologies. Except as expressly set forth above, no other warranties are expressed or implied, including, but not limited to, any implied warranties of merchantability and fitness for a particular purpose, and RTD Embedded Technologies expressly disclaims all warranties not stated herein. All implied warranties, including implied warranties for merchantability and fitness for a particular purpose, are limited to the duration of this warranty. In the event the product is not free from defects as warranted above, the purchaser's sole remedy shall be repair or replacement as provided above. Under no circumstances will RTD Embedded Technologies be liable to the purchaser or any user for any damages, including any incidental or consequential damages, expenses, lost profits, lost savings, or other damages arising out of the use or inability to use the product.

Some states do not allow the exclusion or limitation of incidental or consequential damages for consumer products, and some states do not allow limitations on how long an implied warranty lasts, so the above limitations or exclusions may not apply to you.

This warranty gives you specific legal rights, and you may also have other rights which vary from state to state.

RTD Embedded Technologies, Inc.  
103 Innovation Blvd.  
State College PA 16803-0906  
USA  
Website: [www.rtd.com](http://www.rtd.com)

